

A LOWER EXPONENT OF 1.0418235 FOR PLANAR UNIT DISTANCES

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*This paper was entirely written with AI tools. The listed human author is responsible for having supervised and prompted the AI agents that wrote this paper.

ABSTRACT. Let $u(U)$ count the unordered pairs at distance one in a finite planar set U . Following the number-field construction made explicit by Sawin, we construct sets with $|U_j| \rightarrow \infty$ and

$$\frac{u(U_j)}{|U_j|^{1.0418235}} \longrightarrow \infty.$$

We use mixed-signature fixed fields of complex conjugation in a pro-2 tower with prescribed nonabelian dyadic completion. Averaging norm-one unit orbits cancels the relative regulator and capitulation loss. A complete Hecke-factor calculation in a degree-16384 subfield controls the relative zeta value. Correlated archimedean profiles and weighted finite-place shells give the final improvement. The numerical inequalities are certified by exact finite data and directed interval bounds.

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1. INTRODUCTION

For a finite set $U \subset \mathbb{R}^2$, let

$$u(U) = \#\{\{x, y\} \subset U : |x - y| = 1\}, \quad u(n) = \max_{|U|=n} u(U),$$

where the pairs are unordered and the distance is Euclidean. Erdős introduced the unit distance problem in 1946 [8]. His lattice construction gives

$$u(n) \geq n^{1+c/\log \log n}$$

for some $c > 0$ and arbitrarily large n . The upper bound

$$u(n) = O(n^{4/3})$$

is due to Spencer, Szemerédi and Trotter [34], with a later crossing-number proof due to Székely [35].

The recent number-field construction of OpenAI [29], explained by Alon et al. [1], gives $u(n) \geq n^{1+\delta}$ for a fixed $\delta > 0$ and arbitrarily large n . Sawin [33] made this exponent explicit. In our convention for unordered pairs, his result gives

$$u(n) \geq C^{-1} n^{1.014114}$$

for an absolute constant C and arbitrarily large n . Emmerich [6] obtained a certificate with $\delta = 0.015263\dots$ by optimizing the integer parameters in Sawin's criterion. A subsequent certificate of Emmerich and Cordella [7] reports $\delta = 0.031185334443\dots$. The author's earlier manuscript [26] records $\delta = 0.0358324$. We compare its construction with the present one below.

Our main result is:

Theorem 1.1. *There are finite sets $U_j \subset \mathbb{R}^2$, with $|U_j| \rightarrow \infty$, such that*

$$\frac{u(U_j)}{|U_j|^{1.0418235}} \rightarrow \infty.$$

In particular, $u(n) \geq n^{1.0418235}$ for arbitrarily large integers n .

We obtain the improvement by changing the number fields and counting norm-one unit orbits in mixed signature, then optimizing the local and archimedean overlap in the same construction. The result concerns an unbounded sequence of cardinalities. No generalized Riemann hypothesis or asymptotic class-number equality is used.

1.1. Sawin's Construction. The arithmetic source of unit distances is a quadratic extension K/F with involution ι . If a prime $\mathfrak{p}$ of F splits as $\mathfrak{P}\iota\mathfrak{P}$, then the $k+1$ ideals

$$\mathfrak{P}^j(\iota\mathfrak{P})^{k-j}, \quad 0 \leq j \leq k,$$

have the same relative norm $\mathfrak{p}^k$. Choices at different primes multiply. An ideal-class argument converts some of these choices into elements of a common relative norm. At an embedding where ι acts as complex conjugation, their images have a common length. After rescaling, they give unit displacements in a planar projection of an ideal lattice.

Sawin already uses fractional ideals, relative class numbers, and powers of Frobenius elements to obtain small residue degrees [33]. His explicit construction takes F totally real and K/F CM. A sup-norm window controls the number of vertices and the overlap under unit displacements, while Louboutin's relative class-number bound [20] controls the loss in selecting an ideal class. Prescribing the local behavior requires relations in a pro-2 group. The Golod–Shafarevich criterion [11] ensures that sufficiently few such relations leave an infinite tower. The use of Frobenius cuts to control splitting belongs to the class-field-tower methods of Hajir, Maire and Ramakrishna [12].

There are two related arithmetic precedents. Small split primes and ideal-class selection occur in the work of Ellenberg and Venkatesh [5] on class-group torsion. Number fields with controlled

discriminants have also been used to construct codes and high-dimensional sphere packings [18, 19]. Here the required lattice vectors must have the same length after projection to the plane, which is supplied by their common relative norm.

1.2. The Earlier Bound $\delta = 0.0358324$. The earlier manuscript [26] keeps the totally real/CM setting but changes several estimates in Sawin's criterion. For Euclidean balls of radius $R > 1/2$ and a unit vector e_1 , the overlap has exponential rate

$$\frac{1}{2d} \log \frac{\text{vol}_{2d}(B_R \cap (B_R - e_1))}{\text{vol}_{2d}(B_R)} \longrightarrow \frac{1}{2} \log \left(1 - \frac{1}{4R^2} \right).$$

This replaces the loss $\log(1 - 1/R)$ in the earlier window estimate. A bound for the dual lattice, together with Banaszczyk's transference theorem [2], replaces the packing estimate for the number of vertices by a covolume estimate. The narrow ideal-class calculation also removes an exponential sign loss.

The analytic estimate uses a fixed multiquadratic subfield of every field in the tower. Its normalized Euler product retains more information than a comparison with a power of the Riemann zeta function. This develops the small-prime comparisons in relative class-number estimates [20, 21]. The local construction allows dyadic ramification and charges some relations at degree three in the filtration. For its set of the first 38 odd primes, the displayed infinitude certificate is

$$P(t) = 1 - 38t + 355t^2 + 114t^3, \quad P(6/115) = -\frac{101}{1520875} < 0.$$

These changes, followed by reoptimization, give the earlier manuscript's stated exponent 1.0358324.

Some of these ideas were described in the author's MathOverflow answer [25], which credits spiderduckpig with placing the second split-prime Frobenius relation in filtration degree at least three. Sawin's comment there also records that he knew the fixed-subfield analytic comparison, without having evaluated it. The answer and the archived manuscript are distinct records. In particular, the answer's June 9 change log reports a bookkeeping correction and the figure $\delta > 0.0357$. We use the archived manuscript's 0.0358324 to identify the earlier construction, rather than as a hypothesis of the present proof.

The further improvement comes from allowing a different signature and different local groups. Covolume counting, ball overlap and the removal of the sign loss are already part of the earlier argument. The new counting problem is to use a positive-rank group of norm-one units without losing its regulator.

1.3. Mixed Signature and Norm-One Units. Let $K/\mathbb{Q}$ be totally imaginary and Galois, choose actual complex conjugation ι , and set $F = K^{(\iota)}$. Write

$$[K : \mathbb{Q}] = 2d, \quad [F : \mathbb{Q}] = d = b + 2c, \quad \theta = c/d,$$

where (b, c) is the signature of F . The chosen complex embedding of K restricts to a real embedding of F and supplies the planar projection. The other embeddings need not be real, since ι need not be central. In fact,

$$\frac{b}{d} = \frac{1}{|\text{class}(\iota)|},$$

where the conjugacy class is taken in $\text{Gal}(K/\mathbb{Q})$. Retaining a finite quotient with many conjugates of this involution gives a field whose signature is predominantly complex.

In the CM case, a norm-one element has modulus one at every archimedean coordinate and the norm-one unit group is finite. Above a complex place of F , its two coordinates can instead have moduli e^u and e^{-u} . The norm-one units have rank c and translate these reciprocal pairs. We average over a fundamental domain of their full logarithmic lattice. Tiling gives an exact integral divided by the relative regulator.

For the extensions used here, which are unramified at every finite place, the class-number formula cancels this regulator together with the kernel of extension of ideal classes. More precisely, if $h_{\text{rel}} = h_K/h_F$, κ is the order of that kernel, R_U is the norm-one unit covolume, w_K counts roots of unity, and $\lambda = \text{rd}(K)$, Proposition 5.1 gives

$$\frac{w_K}{h_{\text{rel}}\kappa R_U} = \frac{2^{d-1}\pi^{b+c}}{\lambda^{d/2}L_F(1)}, \quad L_F(s) = \frac{\zeta_K(s)}{\zeta_F(s)}.$$

At $s = 1$, the quotient denotes the quotient of residues. The averaging uses the entire unit lattice, so no independence of its coordinates or equidistribution of unit orbits is needed.

1.4. Local Relations and the Euler Calculation. To construct the fields, we use the filtered images of the local relation modules. Fox derivatives [10] retain the dependencies among relators that a count of their degrees alone would discard. The filtration is the one in the Golod–Shafarevich method, with its group-algebra description [14, 32, 9]. In particular, a relation shared by a global condition and a local condition is subtracted only after identifying the same row in their actual filtered images.

This permits a nonabelian decomposition group of order 32 at 2, with inertia order 8 and residue degree 4. Its relation calculation allows residue degree 2 at both 3 and 5. The resulting family has

$$\lambda = 2^{9/4}\sqrt{15015}, \quad \theta \geq \frac{4095}{8192}.$$

The gain at 3 and 5 has costs. Compared with the preceding five-prime family, the root discriminant increases from $4\sqrt{15015}$, and the residue degree at 17 increases from 2 to 4. The smaller residue degrees also increase the corresponding absolute Euler factors. The exponent improves because the additional norm choices outweigh these losses in the final inequality.

For the analytic calculation we retain two fields

$$N \subset M \subset K, \quad [N : \mathbb{Q}] = 16384, \quad [M : \mathbb{Q}] = 524288.$$

The full Euler calculation is made in N . Its regular representation contains 128 linear, 256 quadratic, 456 quartic and 124 octic irreducible representations, and we evaluate their Hecke factors with the required multiplicities. Further Frobenius information from M is charged only on the remaining primes. This is necessary because adding savings from different subfields can otherwise count the same prime more than once.

The evaluation is at $\sigma = 12001/12000$. The unconditional Tsfasman–Vlăduț prime inequality [41] then gives, for all sufficiently large fields in the tower,

$$\frac{1}{d} \log L_F(1) < 0.042161819.$$

Thus the fixed-subfield idea from the earlier paper is used with a nonabelian field and a complete factor calculation. The finite field identities, conductor data and analytic remainders are proved in sections 3 and 4.

1.5. Archimedean Profiles and Finite Shells. At a complex place of F , a norm-one displacement couples two coordinates through reciprocal scaling. We consequently use a profile on $\mathbb{C}^2$ that depends jointly on their radii. It is a positive cubic Bernstein modification of a Student profile. The coordinates above real places use a Gaussian. Products are taken over places, while the two coordinates at each complex place remain coupled.

At a split finite place of residue-field size Q , a hard valuation window gives the factor

$$(k+1)Q^{-\delta k}.$$

We replace it by a weighted sum of six valuation shells. The shell overlap matrix is explicit, so the gain and the mass can be optimized together. Section 6 gives the exact finite functional.

Both choices enter the same arithmetic and geometric inequality. For hard finite windows its form is

$$\sum_q \frac{\log(k_q + 1) - \delta k_q f_q \log q}{e_q f_q} - (1/2 - \delta) \log \lambda - C + \mathcal{A}_\delta(\theta) > 0,$$

where e_q, f_q are absolute local indices, C bounds $d^{-1} \log L_F(1)$, and $\mathcal{A}_\delta$ is the archimedean contribution. Proposition 5.2 gives the precise expression, and Proposition 6.1 replaces the hard windows by shell weights. Changing the local fields changes several terms at once, which is why the tower and the profiles are optimized together.

The weights are converted into an ordinary finite set by restricting the combined energy to a concentration window. Ideal-class selection uses the resulting overlap weights. A uniform Poisson estimate then bounds the vertices for the same translate and unit deformation that supplies the edges. At

$$\delta = \frac{83647}{2000000} = 0.0418235,$$

the rational witness in section 8 leaves a certified margin greater than $5.33204359816794 \times 10^{-6}$ after the concentration allowance.

1.6. Computation and Formalization. The proof uses finite linear algebra, exact coefficient recurrences, prime data and directed interval enclosures. The arithmetic inputs and the algorithms for checking them accompany the manuscript. Section 8 distinguishes computations rerun by the supplied verifier from stored finite data. These certificates concern numerical inequalities and explicit arithmetic, rather than the statistical accuracy of a floating-point approximation.

There is also a separate Lean development, using Lean 4 and Mathlib [3, 37]. Its selected theorem proves the planar conclusion from one explicit inequality for the ordinary Dedekind zeta function of the fixed field M . That inequality is stated in Remark 4.5. It has not been proved in Lean. The formalization is consequently conditional, and is not used as a substitute for the analytic argument or finite certificate here.

2. THE ARITHMETIC TOWER

We construct an infinite pro-2 extension with prescribed local groups. Its finite subextensions will supply the fields in the geometric and analytic arguments. All pro-2 subgroups and normal closures are topological. For a pro-2 group H , write $D_n H$ for its Zassenhaus filtration, characterized in a finite quotient by $D_n H = \{h : h - 1 \in I_H^n\}$, where I_H is the augmentation ideal of $\mathbb{F}_2[H]$. The associated restricted Lie algebra has bracket induced by commutators and restricted square induced by squaring. Our commutator convention is $[x, y] = x^{-1}y^{-1}xy$. Put

$$T = \{3, 5, 7, 11, 13\}, \quad \Sigma = T \cup \{2\}, \quad E = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{11}, \sqrt{13}).$$

Let $\mathcal{G}$ be the Galois group of the maximal pro-2 extension of $\mathbb{Q}$ unramified at finite primes outside Σ , with no restriction at infinity. Fix actual complex conjugation c .

2.1. The Complete Global Presentation. We use the presentation and cohomological framework for pro- p extensions described in [17, 28]. The odd local relations and the relation at infinity give a complete presentation of the global group.

Lemma 2.1. *The group $\mathcal{G}$ has a minimal presentation with 7 generators and 6 relators. The relators can be chosen to be c^2 and the five odd tame local relators. In particular, the image in the global free group of any dyadic local relation belongs to their closed normal closure.*

Proof. Set $X = \text{Spec } \mathbb{Z}[1/\prod_{p \in \Sigma} p]$. Since 2 is invertible on X , the sheaf μ_2 is constant and Kummer theory gives

$$H^1(X, \mathbb{F}_2) = \langle -1, 2, q : q \in T \rangle_{\mathbb{F}_2}.$$

The displayed squareclasses are independent, so the generator rank is 7.

To bound the relation rank, take the inverse system of connected finite Galois 2-covers $Y \rightarrow X$. Every class in $H^1(Y, \mathbb{F}_2)$ is a quadratic torsor. Its normal closure over X is again a finite Galois 2-cover: its Galois group embeds in an extension of $\text{Gal}(Y/X)$ by a product of copies of C_2 . The direct limit of these first cohomology groups is thus zero. The five-term sequence of the continuous Cartan–Leray spectral sequence gives an injection

$$(2.1) \quad H^2(\mathcal{G}, \mathbb{F}_2) \hookrightarrow H^2(X, \mu_2).$$

We obtain the continuous sequence from finite Galois covers, using the spectral sequence for schemes [22, Theorem 14.9].

The Kummer sequence and $\text{Pic}(X) = 0$ identify the group on the right with $\text{Br}(X)[2]$. Since X is a regular integral arithmetic curve with finite residue fields, the Brauer residue sequence identifies $\text{Br}(X)$ with the classes in $\text{Br}(\mathbb{Q})$ having zero invariant outside $\Sigma \cup \{\infty\}$ [30, Theorem 6.8.3 and Example 6.9.4]. Global Brauer reciprocity then identifies its 2-torsion with the vectors of local 2-torsion invariants supported on $\Sigma \cup \{\infty\}$ whose sum is zero. In particular its dimension is 6. The odd-place and infinite-place coordinates are free, and reciprocity determines the dyadic coordinate [24, Chapter IV, Example 2.14(e)].

To prove surjectivity, we use the quaternion classes obtained by taking cups of global quadratic characters. The class $(-1, -1)$ supplies the infinite coordinate. If $q \equiv 3 \pmod{4}$, then $(-1, q)$ supplies the coordinate at q among the odd places and has zero infinite coordinate. If $(2/q) = -1$, the class $(2, q)$ does the same. The remaining primes of T are 5, 13, and 2 is a quadratic nonresidue at both. These classes span the odd and infinite coordinates. Their only other possible nonzero invariant is at two, whose value is determined by reciprocity.

It follows, by naturality of restriction, that

$$(2.2) \quad H^2(\mathcal{G}, \mathbb{F}_2) \xrightarrow{\sim} \bigoplus_{q \in T} H^2(\mathcal{G}_q, \mathbb{F}_2) \oplus H^2(C_2, \mathbb{F}_2),$$

where $\mathcal{G}_q$ is the maximal pro-2 local group. At odd q , this group has the tame presentation $\langle \tau_q, \phi_q \mid \phi_q \tau_q \phi_q^{-1} = \tau_q^q \rangle$ and a one-dimensional second cohomology group. The real group is C_2 and also has one relation.

For a minimal free pro-2 presentation $1 \rightarrow R \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow 1$, transgression is the natural perfect pairing

$$H^2(\mathcal{G}, \mathbb{F}_2) \simeq \text{Hom}_{\text{cts}}(R/R^2[R, \mathcal{F}], \mathbb{F}_2).$$

This follows from the five-term sequence since $\mathcal{F}$ is free [31, (3.3)–(3.4)]. Lift the local generators to $\mathcal{F}$. Naturality of transgression identifies the dual of (2.2) with the map taking each local defining relator to its class in $R/R^2[R, \mathcal{F}]$. The odd tame relators and c^2 consequently form a basis of this quotient. The lift of c may be chosen as a free generator, because its image is nonzero modulo the Frattini subgroup, detected by $\mathbb{Q}(i)$.

A basis of the relation quotient also gives closed normal generation. Suppose the indicated relators normally generated $R' \neq R$. Some finite 2-quotient of $\mathcal{F}/R'$ would then have nontrivial image N of R/R' . The finite nonzero module $N/\Phi(N)$ has nonzero coinvariants under this finite 2-group: its group algebra has nilpotent augmentation ideal. Hence $N/N^2[N, \mathcal{F}/R'] \neq 0$, contradicting that the indicated relators span $R/R^2[R, \mathcal{F}]$. Finally, a dyadic local relation maps to the identity in $\mathcal{G}$, so its lift belongs to R . $\square$

2.2. The Prescribed Dyadic Field. We choose the following extension at two.

Lemma 2.2. *There is a Galois extension $L_2/\mathbb{Q}_2$ with decomposition group*

$$D = \langle x, y, z \mid x^2, y^2, z^4, [y, z]^2, [x, y], [x, z], [[y, z], z] \rangle$$

of order 32, inertia C_2^3 , residue degree 4, and normalized different exponent $9/4$. Its augmentation Hilbert polynomial is

$$P_D(t) = (1+t)^3(1+t^2)^2,$$

and its socle has degree seven.

Proof. Let $U_4/\mathbb{Q}_2$ be the unramified quartic extension and set

$$A = \mathbb{Q}_2(i, \sqrt{5}, \sqrt{1+2i}), \quad L_2 = A(\sqrt{2})U_4.$$

The norm identity $(1+2i)(1-2i) = 5$ gives the dihedral extension $A/\mathbb{Q}_2$ of order eight. Its maximal abelian subextension is $\mathbb{Q}_2(i, \sqrt{5})$ and its unramified degree is two. Thus $A(\sqrt{2})$ has group $D_8 \times C_2$, degree sixteen, and unramified degree two: its elementary abelian abelianization cannot have a cyclic quotient of order four. Adjoining U_4 doubles the degree and the unramified degree. Its inertia has order eight and is elementary abelian.

Write $w = [y, z]$. The inertia and Frobenius generators can be chosen with the indicated elementary reciprocity classes 5, -1 , -2 . In particular z is a lift of a uniformizer with the specified unit component. The elements x, w are central involutions, y is an involution, and z has order four with z^2 independent of w . These facts give the displayed presentation. Conversely, that presentation makes w central: the identity $w^y = w^{-1}$ and the imposed $w^2 = 1$ give commutation with y , while commutation with z is imposed explicitly. Its normal forms $x^a y^b z^c w^d$, with $a, b, d \in \mathbb{F}_2$ and $c \in \mathbb{Z}/4$, have multiplication

$$(a, b, c, d)(A, B, C, D) = (a + A, b + B, c + C, d + D + cB).$$

These normal forms give precisely thirty-two elements. The two restricted layers have dimensions three and two, which gives the asserted Hilbert polynomial.

For the different, put $k = \mathbb{Q}_2(i)$, $\pi = 1 + i$, and $\alpha = 1 + 2i = 1 + \pi^2$. Here $v_k(\pi) = 1$ and $v_k(2) = 2$. For $b = 1 + \pi$, the element

$$u = \frac{\sqrt{\alpha} - b}{\pi}$$

satisfies the Eisenstein polynomial

$$u^2 + \frac{2b}{\pi}u + \frac{2}{\pi} = 0.$$

It is a uniformizer of $k(\sqrt{\alpha})$ and gives an integral power basis. In its derivative, $2u$ has valuation five and $2b/\pi$ has valuation two. The relative different has exponent two, and the quadratic conductor–discriminant formula gives the same exponent for the character over k . Induction to $\mathbb{Q}_2$ adds $v_2(\text{disc } k) = 2$, so the associated degree-two representation ρ has conductor exponent four. For the twist by χ_2 , dividing 2α by the square π^2 gives $\alpha' = 2 - i$. The element $u' = \sqrt{\alpha'} - 1$ has Eisenstein polynomial

$$(u')^2 + 2u' + 1 - \alpha' = 0,$$

since $v_k(1 - \alpha') = 1$. Its derivative $2(u' + 1)$ has valuation four in the quadratic extension. Thus the relative conductor is four and $\rho \otimes \chi_2$ has conductor exponent six. These calculations use the integral-basis formula for the different and the conductor–discriminant formula for a quadratic character.

The local genus field $E_2 = \mathbb{Q}_2(i, \sqrt{2}, \sqrt{5})$ has discriminant exponent sixteen. The conductor–discriminant identity for $A(\sqrt{2})$ gives

$$v_2(\text{disc}(A(\sqrt{2}))) = 16 + 2(4 + 6) = 36.$$

Its normalized different is $36/16 = 9/4$, which is unchanged by the unramified base extension to L_2 . $\square$

For the choice of generators in this lemma, the elementary reciprocity classes are 5, -1 , -2 , in that order. Here reciprocity classes refer to the maximal elementary abelian quotient. The characters of -1 and 5 on these generators are respectively $y + z$ and z . Consequently the central square form of A is $(y + z)z$, while the unramified quartic contributes z^2 . Their independent sum gives the two central directions $[y, z]$ and z^2 in the presentation. Inertia is $\langle x, y, [y, z] \rangle$. In particular, we may choose the indicated representatives of the two inertia directions to have order two.

The maximal pro-2 group of $\mathbb{Q}_2$ has three generators and one relation. Local Kummer theory gives generator rank three. The pro-2 cohomology comparison in the global proof injects its second cohomology into $\text{Br}(\mathbb{Q}_2)[2]$, and the nonzero local Hilbert pairing makes this injection an isomorphism. The Kummer basis dual to 5, -1 , -2 under that pairing is 2, -5 , 5. Its Hilbert matrix is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

as follows, for odd u, v , from

$$(2^a u, 2^b v)_2 = (-1)^{(u-1)(v-1)/4 + a(v^2-1)/8 + b(u^2-1)/8}.$$

By the relation–cup identity [31, Proposition 3.2], an actual defining relation in these generators has quadratic initial $y^{[2]} + [x, y] + [x, z]$.

2.3. The Global Cuts and Their Finite Quotient. Define G from $\mathcal{G}$ by imposing the full kernel of the local map at two to D , inertia squares at 3, 5, 7, 11, 13, Frobenius squares at 3, 5, and Frobenius fourth powers at 7, 11, 13, 17, 19, 23, 29, 31. At an odd ramified prime choose the Frobenius lift with zero coordinate on its own genus character. We may make this choice by multiplying by inertia. After the inertia square is imposed, the local tame relation makes inertia and Frobenius commute, so the specified even-power cut is independent of this remaining choice. Every cut vanishes in E , so the resulting extension still contains E .

For the finite binary calculations, number the genus basis $(-1, 2, 3, 5, 7, 11, 13)$ by $0, \dots, 6$ and encode $v = \sum v_i e_i$ by the integer $\sum 2^i v_i$. For odd p , let

$$f_p = \sum_{i: a_i \neq p} \frac{1 - (a_i/p)}{2} e_i, \quad (a_0, \dots, a_6) = (-1, 2, 3, 5, 7, 11, 13),$$

where $(\cdot/p)$ is the Legendre symbol. The relevant vectors are

p	3	5	7	11	13	17	19	23	29	31
f_p	43	86	77	83	58	60	71	57	38	101.

The inertia vector at $p \in T$ is $e_{i(p)}$, and the dyadic vectors are $x = 2$, $y = 53$, $z = 89$. Let $S(v)$ and $B(v, w)$ denote the free restricted square and bracket in degree two. In the ordered basis

$$e_0^{[2]}, \dots, e_6^{[2]}, \quad [e_i, e_j] \quad (0 \leq i < j \leq 6)$$

with lexicographically ordered pairs,

$$S(v) = \sum_i v_i e_i^{[2]} + \sum_{i < j} v_i v_j [e_i, e_j], \quad B(v, w) = S(v + w) + S(v) + S(w).$$

The six original quadratic initials are

$$S(e_0), \quad B(e_{i(p)}, f_p) + \frac{p-1}{2} S(e_{i(p)}) \quad (p \in T),$$

with scalars reduced modulo two. These formulas follow by expanding the tame relators and the infinity square. We use them to compute the first three layers of the quotient.

Lemma 2.3. *The actual quotient $Q = G/D_4G$ has order 2^{38} . It retains the entire local group D , local groups C_2^2 at 3, 5, $C_2 \times C_4$ at 7, 11, 13, and C_4 at 17, 19, 23, 29, 31. The conjugacy class of actual complex conjugation in Q has size 2^{12} . Its elementary image is outside every one of these decomposition groups' elementary spans.*

Proof. Use the genus basis $(-1, 2, 3, 5, 7, 11, 13)$. The dyadic generators have elementary vectors

$$x = 2, \quad y = 53, \quad z = 89.$$

The six original odd and infinite quadratic relations are independent, and their sum is $y^{[2]} + [x, y] + [x, z]$. Lemma 2.1 shows that the corresponding original relators give a complete presentation.

The local presentation in Lemma 2.2 has four independent quadratic initials. In its local kernel relation quotient, the actual defining relation r of the maximal pro-2 group of $\mathbb{Q}_2$ has coefficient one on y^2 : its quadratic initial is $y^{[2]} + [x, y] + [x, z]$. The other six local kernel generators have no $y^{[2]}$ coordinate. Replacing y^2 by r preserves generation of the relation quotient, hence closed normal generation by the pro-2 Nakayama argument. Globally r already belongs to the closed normal closure of the six original relators. We need not impose it again.

The resulting complete presentation has sixteen independent quadratic initials: the six originals, five odd inertia squares, $x^{[2]}$, $[x, y]$, $[x, z]$, and the two Frobenius squares at 3, 5. It also has the actual cubic $[[y, z], z]$ and ten degree-four relators, including both z^4 and $[y, z]^2$. Reduction of the local square and commutator forms against these sixteen rows retains exactly the two independent local classes $z^{[2]}$ and $[y, z]$. The three local degree-one generators are independent. Thus G/D_3G , of order 2^{7+12} , already retains all of D . Reduction against the same rows verifies the other displayed local groups and the separation of the elementary image of complex conjugation.

The free restricted cubic layer has dimension 112. Bracketing the sixteen quadratic initials with the seven generators gives rank 92. The actual cubic $[[y, z], z]$ is independent and raises this rank to 93. To see that there are no other degree-three initials, work in the free quotient $\mathcal{F}/D_4\mathcal{F}$. Its second Zassenhaus subgroup is elementary abelian and its third is central. The normal closure of the actual quadratic words ρ_i and actual cubic κ is the vector space spanned by $\rho_i, [\rho_i, X_k], \kappa$. Independence of the sixteen quadratic initials implies that a combination of the ρ_i can lie in degree three only with every coefficient zero. Its intersection with degree three is precisely the span of the displayed brackets and κ . Their initials depend only on the quadratic initials, so unknown cubic coefficients of the actual quadratic words do not change the calculation. Consequently

$$\dim D_3G/D_4G = 19, \quad |Q| = 2^{7+12+19} = 2^{38}.$$

In the preceding 28-coordinate convention, the sixteen rows have hexadecimal masks

$$\begin{aligned} &1, 142104, 1444000, 4480410, \text{a010820}, \text{d020000}, \\ &4, 8, 10, 20, 40, 2, 1\text{a080}, 2\text{c080}, 814\text{aab}, 42\text{aa056}. \end{aligned}$$

We compute the ranks by binary elimination in a free associative algebra. Embed the free restricted degree-two space in the free associative algebra on $X_0, \dots, X_6$ by $e_i^{[2]} \mapsto X_i^2$ and $[e_i, e_j] \mapsto X_iX_j + X_jX_i$. Represent homogeneous cubics by their 343 coefficients on the words $X_iX_jX_k$. For each of the sixteen rows r and each k , insert $rX_k + X_kr$ into a binary row-echelon basis. This produces rank 92. Insert the tensor for $[[y, z], z]$ to obtain rank 93. Running the same operation on all 28 free quadratic basis elements gives rank 112. Ordinary binary elimination in degree two gives rank 16 and verifies independence of the two retained local classes. Reduction of $B(e_0, e_k)$ modulo the sixteen rows gives rank six and kernel $\langle e_0 \rangle$. Reduction of $rX_0 + X_0r$, for the 28 free quadratic basis elements r , modulo the cubic relation space gives rank six. Each calculation uses only the displayed masks and these tensor operations. The supplementary code performs the same integer operations.

Commutation with c has rank six in the second layer and rank six on the second layer with values in the third. Its degree-one kernel is exactly $\langle c \rangle$. Since D_2Q has order 2^{31} and its commutator image

with c has order 2^6 , the intersection of the centralizer with D_2Q has order 2^{25} . Its image in Q/D_2Q is the line through c , giving $|C_Q(c)| = 2^{26}$ and the asserted conjugacy-class size. $\square$

2.4. Filtered Fox Images. We use left Fox coefficients [10]:

$$w - 1 = \sum_i \partial_i w (x_i - 1), \quad \partial_i(uv) = \partial_i u + u \partial_i v.$$

Derivative basis vectors have degree one. If a finite-dimensional space M has a decreasing filtration $F^0 M = M$, eventually zero, write $H_M(t) = \sum_n \dim(F^n M / F^{n+1} M) t^n$. Subspaces have the induced filtration and quotients the quotient filtration. For $0 < t < 1$,

$$H_M(t) = (1 - t) \sum_{n \geq 1} \dim(M / F^n M) t^{n-1}.$$

Consequently a filtration-preserving surjection $M \rightarrow N$ satisfies $H_N(t) \leq H_M(t)$, even without strictness. In particular, for subspaces U, V of a filtered ambient space,

$$(2.3) \quad H_{U+V}(t) \leq H_U(t) + H_V(t) - H_{U \cap V}(t).$$

To obtain this inequality, apply additivity to the quotient filtration from $U \oplus V$, then compare it with the induced filtration on the sum. We next compute the cost of a full local relation kernel.

Lemma 2.4. *Let P be a finite 2-group and $D' \leq P$ one of $C_2, C_4, C_2^2, C_2 \times C_4$, or the group D of Lemma 2.2. Suppose its degree-one directions, and its nonzero degree-two directions when present, embed in the corresponding Zassenhaus layers of P . If D' has r minimal generators and augmentation Hilbert polynomial $P_{D'}$, its full induced local Fox kernel has Hilbert polynomial*

$$(2.4) \quad (rt - 1 + P_{D'}(t)^{-1}) H_{\mathbb{F}_2[P]}(t).$$

Its image under a filtration-preserving map has no larger Hilbert value.

Proof. These local groups have no nonzero layer above degree two, so the hypotheses imply $D' \cap D_n P = D_n D'$ for every n . Extend a homogeneous basis of the graded restricted Lie algebra of D' to one for P , ordering the local factors last. The restricted PBW form of Jennings' theorem [14, 32] gives lifts t_j of the remaining monomials such that

$$\mathbb{F}_2[P] = \bigoplus_j t_j \mathbb{F}_2[D']$$

as right modules and filtered vector spaces. The shift of the j^{th} summand is the weight of t_j , and the shift polynomial is $H_{\mathbb{F}_2[P]}/P_{D'}$. Put $B = \mathbb{F}_2[D']$. The local map $B^r(-1) \rightarrow I_{D'}$, $(a_i) \mapsto \sum a_i(d_i - 1)$, is strict, because $I_{D'}^n = \sum_i I_{D'}^{n-1}(d_i - 1)$. Its kernel has polynomial $rtP_{D'} - (P_{D'} - 1)$. Filtered right-freeness preserves exactness and strictness under induction, proving the formula. Finally, the local generator words have independent elementary images in a minimal global free presentation and so extend to a free pro-2 basis. The Fox change-of-basis matrix and its inverse preserve augmentation degree. Thus the identification with a coordinate submodule in the global derivative module preserves the induced filtration. The assertion about images follows from the cumulative Hilbert formula. $\square$

The dyadic block shares a relation with the odd and infinite blocks. We compute the image of this relation as well as the full dyadic cost.

Lemma 2.5. *For this dyadic local group the full induced local Fox-image cost is $d_D(t) = 3t - 1 + P_D(t)^{-1}$. The common actual dyadic defining row has normalized image polynomial*

$$s_D(t) = t^2 \left(1 - \frac{t^7}{P_D(t)} \right).$$

Proof. The independent local degree-one and degree-two classes in Lemma 2.3 give an embedding of the local restricted graded Lie algebra in the global one. Extend its restricted PBW basis to a global basis, with the local factors last. This proves filtered freeness as a right local-algebra module, including the exact augmentation shifts. The local augmentation sequence then gives the full Fox-image cost $3t - 1 + P_D^{-1}$ by Lemma 2.4. This argument does not require commutativity.

Write $B = \mathbb{F}_2[D]$. The linear coefficients of the actual local relation's Fox row span I_B/I_B^2 , because the local Hilbert pairing is nonsingular [24, Chapter III, Theorem 4.4], and its matrix is the matrix of linear Fox coefficients [31, Proposition 3.2]. In our coordinates the relation has initial $y^{[2]} + [x, y] + [x, z]$, whose three derivative coefficients are $y + z, y + x, x$ and are independent. They generate I_B as a right ideal by Nakayama. Their common left annihilator is the one-dimensional left socle of B . We can check the required graded assertion in this 32-element algebra. Starting with its displayed normal-form multiplication, form $J^0 = B$ and obtain J^{n+1} by spanning the right products of a basis of J^n by $x - 1, y - 1, z - 1$. The dimensions for $0 \leq n \leq 8$ are

$$32, 31, 28, 23, 16, 9, 4, 1, 0.$$

On J^n/J^{n+1} the simultaneous multiplication map to $(J^{n+1}/J^{n+2})^3$ has kernel zero for $n < 7$ and dimension one for $n = 7$. This is a finite row reduction on vectors of length 32 using the specified multiplication. The top vector is the sum of all group elements. Changing the three degree-one coefficients by an invertible matrix does not change these kernels. Thus every nonzero homogeneous element below degree seven has nonzero product with at least one linear Fox coefficient. Row multiplication raises coefficient degree by exactly one, and the derivative basis shift contributes one more. The row image consequently has polynomial $t^2(P_D - t^7)$. Filtered induction gives the stated normalized cost.

The actual row belongs to the full dyadic block because the local relation vanishes in D . It also belongs to the sum of the odd blocks and infinity square. Indeed, by the complete global presentation, the local relation belongs to the closed normal closure of those original relators. If a defining relator ρ evaluates to one in the finite group, the Fox product rule gives

$$\partial(upu^{-1}) = u\partial\rho.$$

Products of conjugates give module combinations of the original rows. Evaluation in the finite derivative module is continuous and its row span is closed, so the same conclusion holds for the closed normal closure. The filtered local coordinate change used above preserves the computed image polynomial. Thus the common submodule is an actual intersection in the global derivative module. $\square$

2.5. Infinitude and the Resulting Number Fields. The local Fox bounds imply that the global group is infinite and give the following family of number fields.

Theorem 2.6. *The group G is infinite. For every sufficiently large power of two d there is a totally imaginary Galois extension $K/\mathbb{Q}$ of degree $2d$ retaining $Q = G/D_4G$. For its actual complex conjugation c , the fixed field $F = K^{\langle c \rangle}$ has at least one real embedding and*

$$\text{rd}(K) = \text{rd}(F) = 2^{9/4}\sqrt{15015}, \quad \frac{4095}{8192} \leq \theta := \frac{r_2(F)}{d} < \frac{1}{2}.$$

The extension K/F is quadratic, unramified at every finite place, and split at every place above the following rational primes, whose uniform local types in both K and F are

p	e_p	f_p
2	8	4
3, 5	2	2
7, 11, 13	2	4
17, 19, 23, 29, 31	1	4.

Proof. Suppose first that G is finite and put $A = \mathbb{F}_2[G]$. The strict global augmentation map $A^7(-1) \rightarrow I_G$ has kernel L with $H_L = 1 - (1 - 7t)H_A$. The continuous Fox relation sequence says that the rows of the complete defining relations generate L . This also follows by tensoring the free augmentation sequence and using the ideal normally generated by the defining relators. The local hypotheses of Lemma 2.4 hold by Lemma 2.3. Write

$$c_1 = \frac{t^2}{1+t}, \quad c_2 = 2t - 1 + (1+t)^{-2}, \quad c_{24} = 2t - 1 + \frac{1}{(1+t)^2(1+t^2)}, \quad c_4 = \frac{t^4}{(1+t)(1+t^2)}.$$

The two odd C_2^2 blocks cost $2c_2H_A$, the three odd $C_2 \times C_4$ blocks cost $3c_{24}H_A$, infinity costs c_1H_A , and the five unramified order-four caps cost $5c_4H_A$. The full dyadic block costs d_DH_A . Its actual intersection with the sum of the odd and infinite blocks contains the common row image s_DH_A , which is subtracted once by (2.3). Consequently

$$1 \leq H_A(t)\mathcal{P}(t), \quad \mathcal{P}(t) = 1 - 7t + 2c_2 + 3c_{24} + d_D + c_1 - s_D + 5c_4.$$

Exact rational substitution gives

$$\mathcal{P}(11/34) = -\frac{160218343}{112275691650} < 0.$$

This contradicts finiteness. The inequality includes every defining relation, and the subtracted term comes from the common row image in the global Fox kernel.

An infinite profinite group has finite quotients of unbounded order. Taking common refinements with Q gives arbitrarily large finite 2-quotients $H \twoheadrightarrow Q$. A nontrivial kernel meets $Z(H)$ nontrivially by the class equation for conjugation, and hence contains a central subgroup of order two. Repeated quotienting by such subgroups inside the kernel realizes every intermediate power-of-two order while retaining Q . These quotients correspond to the claimed Galois fields.

Because K contains i , it is totally imaginary. The chosen complex conjugation fixes a real subfield in the given embedding, so F has a real embedding. If $b = r_1(F)$ and $H = \text{Gal}(K/\mathbb{Q})$, embeddings of F correspond to cosets of $\langle c \rangle$ in H . A coset is real exactly when its representative centralizes c . Hence

$$b = |C_H(c)|/2, \quad \frac{b}{d} = \frac{|C_H(c)|}{|H|}.$$

The conjugacy class in H maps onto the class in Q , of size 2^{12} . Thus $0 < b/d \leq 2^{-12}$, giving the stated interval for $\theta = (1 - b/d)/2$.

Conjugation preserves an elementary image. The separation verified in Lemma 2.3 excludes every conjugate of c from each ramified or selected decomposition group. Intersection with $\langle c \rangle$ is trivial, proving relative splitting at these places. Elsewhere $K/\mathbb{Q}$ is unramified, so K/F is unramified there also. Relative splitting transfers the displayed local types to F . Tame inertia of order two contributes $p^{1/2}$ to the root discriminant at each $p \in T$, and the dyadic lemma contributes $2^{9/4}$. Finally $\Delta_K = \Delta_F^2$ because the relative finite discriminant is one. $\square$

3. AN EXPLICIT RETAINED FIELD AND ITS HECKE FACTORS

The analytic estimate uses a fixed subfield of every sufficiently large field in Theorem 2.6. We construct this subfield by Kummer theory and express its zeta function as a product of Dirichlet and Hecke functions. Exact local calculations then determine the conductors and Euler polynomials needed to evaluate these factors.

3.1. A Catalog of Norm Extensions. Retain $V = \text{Gal}(E/\mathbb{Q}) = \mathbb{F}_2^7$ with the coordinates of Section 2. For a rational squareclass A supported on $-1, 2, 3, 5, 7, 11, 13$, write $\chi_A \in V^*$ for its

Kummer character. Define $\alpha_i = x_i + y_i\sqrt{A_i}$ by the following seventeen rows. Each row satisfies $N_{\mathbb{Q}(\sqrt{A_i})/\mathbb{Q}}(\alpha_i) = B_i z_i^2$.

$$(3.1) \quad \begin{array}{c|c|c|c|c|c} i & A_i & B_i & x_i & y_i & z_i \\ \hline 0 & -1 & 65 & 4 & 7 & 1 \\ 1 & -55 & 91 & 6 & 1 & 1 \\ 2 & -2 & 33 & 1 & 4 & 1 \\ 3 & -55 & 14 & 1 & 1 & 2 \\ 4 & -55 & 26 & 7 & 1 & 2 \\ 5 & -7 & 2 & 1 & 1 & 2 \\ 6 & -10 & 1001 & 1 & 10 & 1 \\ 7 & 22 & 273 & 19 & 2 & 1 \\ 8 & -39 & 55 & 4 & 1 & 1 \\ 9 & -35 & 429 & 1 & 7 & 2 \\ 10 & -143 & 42 & 5 & 1 & 2 \\ 11 & -3 & 13 & 1 & 2 & 1 \\ 12 & -26 & 105 & 1 & 2 & 1 \\ 13 & -6 & 385 & 1 & 8 & 1 \\ 14 & -14 & 65 & 3 & 2 & 1 \\ 15 & -11 & 5 & 3 & 1 & 2 \\ 16 & -1 & 13 & -2 & 3 & 1 \end{array}$$

All these numbers are integral. Their base discriminants and norms are supported on $\Sigma = \{2, 3, 5, 7, 11, 13\}$, so their Galois closures are unramified outside Σ . An individual catalog field need not survive all the cuts in G . We will check retention for the products used in the construction.

For a nonnegative integer $a < 2^{17}$ define

$$\beta(a) = \prod_{i=0}^{16} \alpha_i^{a_i}, \quad q_a(v) = \sum_{i=0}^{16} a_i \chi_{A_i}(v) \chi_{B_i}(v), \quad a = \sum_i 2^i a_i.$$

Products and signs in the second formula are taken in $\mathbb{F}_2$. The norm identities determine the central extension associated to each product.

Lemma 3.1. *For any catalog word a , the field $E(\sqrt{\beta(a)})$ is Galois over $\mathbb{Q}$, and its invariant Kummer class has transgression q_a . When this form is nonzero, the field is a central extension of V by C_2 with that quadratic form. Products of the catalog radicands add their extension classes. On the space of invariant Kummer classes in the maximal pro-2 Σ -unramified extension above E , this class map is injective.*

Proof. Choose the square roots of the positive rational numbers B_i compatibly inside E . If $v \in V$ changes the sign of $\sqrt{A_i}$, put

$$u_{i,v} = \frac{z_i \sqrt{B_i}}{\alpha_i}.$$

Otherwise put $u_{i,v} = 1$. The norm identity gives

$$\alpha_i u_{i,v}^2 = v(\alpha_i), \quad u_{i,v} v(u_{i,v}) = (-1)^{\chi_{A_i}(v) \chi_{B_i}(v)}.$$

Thus v lifts to an automorphism by sending $\sqrt{\alpha_i}$ to $u_{i,v} \sqrt{\alpha_i}$. For a product radicand use $u_v = \prod_i u_{i,v}^{a_i}$. It satisfies

$$(3.2) \quad \beta(a) u_v^2 = v(\beta(a)), \quad u_v v(u_v) = (-1)^{q_a(v)}.$$

The kernel over E is central of exponent two. The squares of lifted elements are given by q_a , and their commutators by its polar form $B_{q_a}(v, w) = q_a(v + w) + q_a(v) + q_a(w)$. These squares and commutators determine the central extension, since every word can be collected into an ordered

product of the seven lifted basis elements and a central sign. Applying the same calculation to a product of radicands gives the addition rule.

For injectivity let $\mathcal{H}$ be the subgroup of $\mathcal{G}$ fixing E . The five-term sequence contains

$$H^1(\mathcal{G}, \mathbb{F}_2) \longrightarrow H^1(\mathcal{H}, \mathbb{F}_2)^V \xrightarrow{\text{trg}} H^2(V, \mathbb{F}_2).$$

Every quadratic character of $\mathcal{G}$ factors through E , by the Kummer computation in Lemma 2.1. The first arrow is zero, so transgression is injective. The calculation of squares and commutators above identifies this transgression with the indicated quadratic form. In particular a nonzero form cannot come from a square radicand in E , and independent forms give independent Kummer classes. $\square$

3.2. The Completed Field and the Full Quadratic Quotient. Define two lists of raw catalog words

$$(3.3) \quad \mathcal{A} = (17, 27184, 6220, 2, 32768, 39508, 132),$$

$$(3.4) \quad \mathcal{B} = (2, 256, 512, 4096, 9, 65, 132, 2052, 24576, 37, 32768, 65536).$$

Set

$$N = E(\sqrt{\beta(a)} : a \in \mathcal{A}), \quad M = E(\sqrt{\beta(b)} : b \in \mathcal{B}).$$

For example, writing $\eta = \alpha_{13}\alpha_{14}$, the seven defining radicands of N are

$$\begin{aligned} \gamma_1 &= \alpha_0\alpha_4, & \gamma_2 &= \alpha_4\alpha_5\alpha_9\alpha_{11}\eta, \\ \gamma_3 &= \alpha_2\alpha_3\alpha_6\alpha_{11}\alpha_{12}, & \gamma_4 &= \alpha_1, \\ \gamma_5 &= 3 + \sqrt{-11}, & \gamma_6 &= \alpha_2\alpha_4\alpha_6\alpha_9\alpha_{11}\alpha_{12}\gamma_5, \\ \gamma_7 &= (1 + 4\sqrt{-2})(19 + 2\sqrt{22}). \end{aligned}$$

These products give two subfields of the tower.

Proposition 3.2. *The fields N and M are contained in every field K of Theorem 2.6. Their degrees are*

$$[N : \mathbb{Q}] = 2^{14} = 16384, \quad [M : \mathbb{Q}] = 2^{19}, \quad N \subset M.$$

The field M is exactly the field corresponding to G/D_3G .

Proof. Use the 28-coordinate convention dual to the restricted quadratic basis in Section 2: the first seven coordinates encode v_i^2 and the remaining 21 encode $v_i v_j$ for lexicographically ordered $i < j$. In this convention the forms for $\mathcal{A}$ and $\mathcal{B}$ have hexadecimal masks

$$(3.5) \quad \begin{array}{l|l} \mathcal{A} & 9014280, \text{af}24500, 6830680, \text{b}401400, \\ & 800200, \text{acf}0980, \text{a}138900 \\ \mathcal{B} & \text{b}401400, 9140\text{a}00, 78\text{c}1900, 520\text{e}700, \\ & 2415680, 1\text{c}38\text{e}00, \text{a}138900, 213900, \\ & 45\text{f}9\text{c}00, 1\text{b}\text{b}80, 800200, 1000. \end{array}$$

To obtain these masks, expand each product of linear characters from (3.1), writing a diagonal term as v_i^2 . Binary elimination gives ranks seven and twelve, respectively, and shows that the first span is contained in the second. Taking the binary scalar product with each of the sixteen displayed relation rows in Section 2 gives zero for every form in the second list. The rows have rank 16 in dimension 28, so these twelve independent forms span their entire annihilator.

Lemma 3.1 now proves the asserted Kummer degrees and the containment. Each extension has central kernel of exponent two over the exponent-two group V , and hence class at most two and exponent at most four. Its third Zassenhaus subgroup is thus trivial. Evaluating a defining relator of G in this group depends only on its quadratic initial, and the higher-degree relators vanish. Since the forms annihilate every quadratic relation, these number fields satisfy all the defining relations of G , including the local cuts. Finally G/D_3G has the same order 2^{19} as M , by Lemma 2.3, which proves equality. Since the fields K retain G/D_4G , they retain both M and N . $\square$

3.3. The Complete Representation Inventory. For a quadratic form q on V , write $R_q = \text{rad}(B_q)$ and $g(q) = \sum_{v \in V} (-1)^{q(v)}$. The space of forms defining N will be denoted $\mathcal{Q} = \langle q_a : a \in \mathcal{A} \rangle$. To count the irreducible representations and their gamma factors, we examine the polar forms and the action of complex conjugation.

Lemma 3.3. *The irreducible representations of $\text{Gal}(N/\mathbb{Q})$ comprise 128 of degree one, 256 of degree two, 456 of degree four, and 124 of degree eight. Exactly eight of the degree-four representations are pure at infinity: four have gamma vector $(0, 0, 0, 0)$ and four have $(1, 1, 1, 1)$. Every other nonlinear representation is mixed, with equal numbers of zeros and ones in its gamma vector.*

Proof. Fix a nontrivial character of $\text{Gal}(N/E)$ and let $q \in \mathcal{Q} \setminus \{0\}$ be its central form. Suppose $\text{rank } B_q = 2m$. The corresponding central pushout has commutator pairing B_q , and its center above V is the inverse image of R_q . There are $|R_q| = 128/2^{2m}$ central characters extending the chosen nontrivial kernel character. Each gives one irreducible representation of dimension 2^m . To construct these representations, choose a maximal isotropic subspace of V for B_q , of dimension $7 - m$ and containing R_q . Its inverse image is abelian. Extend the chosen central character to it and induce to the full group. Nondegeneracy of the induced pairing on V/R_q shows, by the usual intertwiner criterion for induction, that no element outside that subgroup intertwines its character. The induced representation is irreducible and has degree 2^m . Changing the extension inside this subgroup only conjugates the character, while changing its restriction to the center gives distinct representations. Their squared dimensions sum to 128, so this accounts for the entire central sector. The trivial central character gives the 128 linear characters of E .

Enumeration of the 127 nonzero sums of the seven masks in (3.5) gives

rank B_q	2	4	6
$\#q$	8	57	62
$\#\{q : g(q) > 0\}$	5	34	48
$\#\{q : g(q) = 0\}$	3	23	14.

The positive sums are respectively 64, 32, and 16. No negative Gauss sum occurs. For this enumeration, evaluate each quadratic polynomial on the 128 binary vectors and compute the rank of its 7 by 7 polar matrix. The resulting dimension count is

$$128 + 256 \cdot 2^2 + 456 \cdot 4^2 + 124 \cdot 8^2 = 16384.$$

The linear map $q \mapsto B_q(c, \cdot)$ on $\mathcal{Q}$, where $c = e_0$, has rank six. Its only nonzero kernel form has rank four and is represented by $\alpha_4\alpha_6\alpha_9\alpha_{10}\alpha_{11}$. Every form satisfies $q(c) = 0$, as it annihilates the infinity square. If $B_q(c, \cdot) \neq 0$, an element conjugating c to its product with the nontrivial central sign shows that the trace of c in the representation is zero. Because $c^2 = 1$, its $+1$ and -1 eigenspaces then have equal dimensions. For the unique remaining sector c is central in the pushout, hence acts scalarly. Its eight genus twists split equally between the two signs. The explicit description below identifies the even and odd twists. The dimensions of these eigenspaces give the asserted real gamma factors. $\square$

The regular representation gives the exact Artin factorization

$$(3.6) \quad \zeta_N(s) = \prod_{\rho \in \text{Irr}(\text{Gal}(N/\mathbb{Q}))} L(s, \rho)^{\dim \rho}.$$

This follows prime by prime by decomposing the regular representation and taking the determinant on inertia invariants. We next identify every nonlinear factor with an entire Hecke function, which allows us to use this identity without an Artin holomorphy assumption.

3.4. Descent to Quadratic Hecke Characters. We construct the inducing characters by descending the catalog radicands to suitable subfields of E . The descent gives exact identities for the Artin functions in the analytic calculation.

If $g(q) > 0$ and $\text{rank } B_q = 2m$, then q vanishes on its radical and the induced nondegenerate quadratic form has Arf invariant zero. Equivalently it has a totally singular subspace U of dimension $7 - m$ containing R_q . We may find such a subspace by enumerating the subspaces of V and testing $q(u) = 0$ on each vector. Put $B = E^U$, so $[B : \mathbb{Q}] = 2^m$, and choose a catalog product β representing q . Let u_v be the exact lifts from (3.2). On E define $T_v P = u_v v(P)$. For $v \in U$ these are commuting involutions: their squares and commutators are respectively $(-1)^{q(v)}$ and $(-1)^{B_q(v,w)}$.

Starting from $P = 1$, take a basis $v_1, \dots, v_{7-m}$ of U and successively replace

$$(3.7) \quad P \mapsto P + T_{v_j} P \quad \text{or, if this is zero,} \quad P - T_{v_j} P.$$

At least one choice is nonzero in characteristic zero. Commutativity preserves the eigenspaces already selected, so the final nonzero P satisfies $T_v P = \pm P$ for every $v \in U$. Consequently

$$\eta = \beta P^2 \in B, \quad E(\sqrt{\eta}) = E(\sqrt{\beta}).$$

Every operation is rational arithmetic in the basis $\prod_i \sqrt{a_i^{\epsilon_i}}$ of E . At a step, if $\eta_0 = \beta P^2$, the element $h = \beta u_v P v(P)$ satisfies

$$h^2 = \eta_0 v(\eta_0), \quad \beta(P \pm T_v P)^2 = \eta_0 + v(\eta_0) \pm 2h.$$

At each descent step, these identities verify the new radicand, while the nonzero test and the invariance under the processed subgroup verify the chosen projector.

Let $L = B(\sqrt{\eta})$ and let λ be its nontrivial quadratic Hecke character over B . The inverse image of U in the central pushout is the elementary abelian subgroup used in the representation proof. Its character selecting $\sqrt{\eta}$ induces the required irreducible representation. Thus

$$(3.8) \quad L(s, \rho) = L_B(s, \lambda) = \frac{\zeta_L(s)}{\zeta_B(s)}.$$

This is an entire nontrivial Hecke function. Its completed functional equation has root number $+1$, because the completed Dedekind zeta functions in the quotient have root number $+1$. The remaining representations in this sector are obtained by multiplying η by rational squareclasses D representing $V^*/B_q(V)$. There are $|R_q|$ such twists. Here $B_q(V)$ denotes the image of $v \mapsto B_q(v, \cdot)$. Twisting by a character in this image conjugates the inducing representation, while the quotient distinguishes its central characters.

If $g(q) = 0$, choose $p \in \{5, 13\}$ so that $g(q + \chi_p) > 0$. For this seven-dimensional space, exact Gauss-sum evaluation gives 26 sectors permitting either choice, nine requiring $p = 5$, and five requiring $p = 13$. Thus a suitable choice exists in every sector. Put

$$\beta_5 = -10 - 2\sqrt{5}, \quad \beta_{13} = -26 - 6\sqrt{13}.$$

Their norms are $16p$ and their central forms are χ_p . To identify these as the cyclic quartic extensions of conductor p , we use their defining polynomials

$$f_5(X) = X^4 + 20X^2 + 80, \quad f_{13}(X) = X^4 + 52X^2 + 208.$$

If $r^2 = \beta_p$, the conjugation rule $r \mapsto 4\sqrt{p}/r$ has square $r \mapsto -r$, and gives their cyclic Galois group of order four. The following integral bases have trace-pairing discriminants 5^3 and 13^3 , respectively:

$$p = 5: \quad 1, \quad r^2/8 - r/4 + 3/2, \quad -r^2/8 - r/4 - 3/2, \quad r^3/16 - r^2/8 + 3r/4 - 1;$$

$$p = 13: \quad 1, \quad r^2/24 - r/4 + 5/6, \quad -r^2/24 - r/4 - 5/6, \quad r^3/48 - r^2/24 + 11r/12 - 4/3.$$

Integrality and the discriminants follow by substitution in f_p and rational trace-matrix arithmetic. These orders are maximal. The field is unramified outside p , and its quadratic subfield $\mathbb{Q}(\sqrt{p})$ is ramified at p . Hence inertia is the full cyclic group of order four. Tame ramification gives discriminant

exponent three, which agrees with the computed order discriminant. The conductor–discriminant formula gives conductor p for both faithful quartic characters, identifying them with ξ_p and $\bar{\xi}_p$ under global reciprocity. Descend $\beta\beta_p$ by the positive construction. If ρ' is its quadratic induction and ξ_p is the primitive quartic Dirichlet character normalized by $\xi_p(2) = i$, use $\rho' \otimes \xi_p$. The two quadratic forms add to $(q + \chi_p) + \chi_p = q$, so this is a representation of the actual original central field. This can also be seen directly on the two Kummer square roots: the extra central sign from β_p occurs in both factors and cancels in their tensor product. Its genus twists again give every representation in the sector. Since $\bar{\xi}_p = \xi_p\chi_p$ and ρ' is real, the genus twists pair the complex representation with its conjugate. The resulting functions are entire Hecke functions with root numbers of modulus one. The bounds below do not require their phases to be known.

3.5. The Pure Quartic Field. The unique pure form in Lemma 3.3 is represented by $\beta = \alpha_4\alpha_6\alpha_9\alpha_{10}\alpha_{11}$. Five projector steps give

$$B_0 = \mathbb{Q}(\sqrt{30}, \sqrt{65}), \quad \eta_0 = 393911 - 10360\sqrt{30} + 23013\sqrt{65} - 2400\sqrt{1950}.$$

To verify this squareclass identity, start with $P_0 = 1$ and the multipliers u_v for this β . Apply $P_j = P_{j-1} + \epsilon_j T_{v_j} P_{j-1}$ with

$$(v_1, \dots, v_5) = (1, 16, 32, 6, 74), \quad (\epsilon_1, \dots, \epsilon_5) = (1, 1, 1, -1, -1).$$

Exact multiplication in E gives $\eta_0 = \beta P_5^2/256$. The five vectors span the simultaneous kernels of χ_{30} and χ_{65} , so their fixed field is B_0 . All four real values are positive: using $\sqrt{30} < 6$, $\sqrt{65} < 9$, and $\sqrt{1950} < 45$, each is larger than $393911 - 10360 \cdot 6 - 23013 \cdot 9 - 2400 \cdot 45 = 16634$. The eight factors are

$$(3.9) \quad L_D(s) = \frac{\zeta_{B_0(\sqrt{D}\eta_0)}(s)}{\zeta_{B_0}(s)}, \quad D \in \{\pm 1, \pm 2, \pm 3, \pm 5\}.$$

The base discriminant is $120 \cdot 65 \cdot 312 = 2433600$. For every listed D , the maximal-order discriminant of $B_0(\sqrt{D}\eta_0)$ is

$$2^{14}3^45^47^211^213^4 = 140455850895360000.$$

The ratio is 240240^2 , giving the conductor of each factor by the conductor–discriminant formula. Positive D have four even gamma factors, negative D four odd ones, and every root number is $+1$. The discriminants here are those of maximal orders, as required by the formula below.

3.6. Exact Local Data. The inducing characters determine the completed functions used in the analytic estimate.

Proposition 3.4. *All irreducible factors of ζ_N are the actual Dirichlet factors and Hecke inductions just constructed. The gamma inventory is the one in Lemma 3.3. Their conductors and exceptional Euler denominators are given by the following finite local algorithms, with exact inducing-field, radicand, twist, and projector data in the supplement. In particular no conjectural Artin holomorphy, conductor identification from a truncated Euler product, or guessed root number is used.*

Proof. The preceding constructions give every irreducible representation in the regular representation. To compute its Hecke data in a positive sector with twist D , compute the maximal orders of B and $L_D = B(\sqrt{D}\eta)$. The conductor–discriminant formula gives

$$(3.10) \quad Q_\rho = \frac{|\Delta_{L_D}|}{|\Delta_B|} = |\Delta_B| N_{B/\mathbb{Q}}(\mathfrak{o}_{L_D/B}).$$

For a prime p , factor p in both maximal orders. With $f(\mathfrak{p}/p)$ denoting residue degrees, put

$$D_{B,p}(T) = \prod_{\mathfrak{p}|p} (1 - T^{f(\mathfrak{p}/p)}), \quad D_{L_D,p}(T) = \prod_{\mathfrak{P}|p} (1 - T^{f(\mathfrak{P}/p)}).$$

The exact local denominator of (3.8) is

$$(3.11) \quad D_{\rho, \mathfrak{p}}(T) = D_{L_D, \mathfrak{p}}(T) / D_{B, \mathfrak{p}}(T).$$

This quotient is a polynomial: above each prime of B , quadratic splitting contributes $1 - T^f$, inertia degree two contributes $1 + T^f$, and ramification contributes 1. Its degree is at most $\dim \rho$. At a real place its character is even or odd according as $D\eta$ is positive or negative. At a complex place the induction contributes one even and one odd real gamma factor.

We can also compute these data locally, including in the sectors with a quartic twist. At an odd prime of B , first remove the even valuation of the quadratic radicand by a square. An odd remaining valuation is ramified with conductor exponent one. For valuation zero, the residue quadratic character determines splitting, giving the unramified value of the quadratic character on a uniformizer. At a dyadic place, use the finite unit quotient and square test: a unit $1 + u$ with valuation $v(u) > 2v(2)$ is a square by Hensel's lemma. Thus squareclass and quadratic-character computations reduce to a finite quotient of the local integer ring. Its ramification and conductor can be determined by testing the character on the descending unit groups. The quadratic norm criterion determines each character value. All these tests use exact finite-ring arithmetic.

For $\rho = \text{Ind}_B^{\mathbb{Q}} \lambda \otimes \xi_p$, apply these tests to the local character

$$\lambda_{\mathfrak{p}}(\xi_{p, \ell} \circ N_{B_{\mathfrak{p}}/\mathbb{Q}_{\ell}}) \quad (\mathfrak{p} \mid \ell).$$

The local component of a primitive Dirichlet character is normalized to have the usual Frobenius value away from its modulus. At its modulus, its restriction to units is the inverse of the finite Dirichlet character, and its value on the rational uniformizer is one. For each local character, its conductor exponent is the least integer n for which it is trivial on $1 + \mathfrak{p}^n$, with exponent zero for an unramified character. The search is finite: its conductor is no larger than the maximum of the two known factor conductors. Induction then gives

$$a_{\ell}(\rho) = \sum_{\mathfrak{p} \mid \ell} f(\mathfrak{p}/\ell) (a_{\mathfrak{p}}(\lambda \xi_p \circ N) + d_{B_{\mathfrak{p}}/\mathbb{Q}_{\ell}}),$$

where $d_{B_{\mathfrak{p}}/\mathbb{Q}_{\ell}}$ is the exponent of the local different. If a local inducing character is ramified it contributes no Euler factor. If it is unramified, with uniformizer value ε , it contributes $1 - \varepsilon T^{f(\mathfrak{p}/\ell)}$. Multiplication over $\mathfrak{p} \mid \ell$ supplies the target denominator with its actual complex phase.

At the cancelling odd prime this calculation has a particularly short form. Write a local uniformizer as $\pi^2 = pr$, let u be the normalized quadratic radicand unit, and write f for the residue degree. The cancelled character is unramified and its value is

$$(3.12) \quad \varepsilon = (u/\mathbb{F}_{p^f}) \xi_p((-r)^f)^{-1}.$$

The first factor is the value of the unramified quadratic character, and the second comes from the unit part of the norm of π . The inverse is required by the local normalization at the modulus. This determines each complex Euler polynomial separately.

The finite arithmetic records specify the rational defining data, projector identities, primitive character values, and exceptional polynomials. The supplementary checks apply the displayed identities and local operations to these records. Independent maximal-order and prime-factorization computations check the discriminant ratios and local polynomials a second time. The identification of the infinite Euler products follows from the inducing characters constructed above. $\square$

3.7. Frobenius Tests for the Prime Budget. For an ordinary prime $p \notin \Sigma$, let $v_p \in V$ be its genus Frobenius vector. If $v_p \neq 0$, a lift in either central field has square measured by the corresponding quadratic forms. Consequently

$$(3.13) \quad \begin{aligned} f_N(p) = 4 &\iff q_a(v_p) = 1 \text{ for some } a \in \mathcal{A}, \\ f_M(p) = 4 &\iff q_b(v_p) = 1 \text{ for some } b \in \mathcal{B}. \end{aligned}$$

If no form detects the square, the residue degree is two, since the genus vector is nonzero. In particular a prime with $f_N(p) = 2$ and $f_M(p) = 4$ has residue degree at least four in every field K .

If $v_p = 0$, choose any square root s_i of A_i modulo p and set

$$\varepsilon_i(p) = \left(\frac{x_i + y_i s_i}{p} \right) \in \{\pm 1\}.$$

The norm identity shows both that the argument is nonzero and that the sign is independent of the choice of s_i : the product with its conjugate is the nonzero square $B_i z_i^2$ modulo p . For a raw word a , its central Frobenius sign is $\prod_i \varepsilon_i(p)^{a_i}$. The prime splits completely in N exactly when every word in (3.3) has sign $+1$. The analogous test for M uses (3.4). Any detected central sign gives residue degree two in that field. Since $N \subset M$, the completed-field detections are a subset of the full-field detections. Order the seven completed predicates before the twelve full predicates, and assign each prime to its first nonzero predicate. This gives disjoint sets of primes detected in the two fields, which we will use for the finite prime corrections in the analytic estimate.

4. THE RELATIVE ZETA FACTOR AND ITS EXPLICIT CEILING

We bound the relative zeta value needed in the geometric construction. Each zeta function is normalized by the absolute degree of its field. With this normalization, the relative degree contributes no additional factor of two to the final bound.

Let K be a field in Theorem 2.6, let c_0 be its actual complex conjugation, and put

$$F = K^{\langle c_0 \rangle}, \quad [K : \mathbb{Q}] = 2d, \quad [F : \mathbb{Q}] = d = b + 2c, \quad \theta = c/d.$$

The extension K/F is quadratic and unramified at all finite primes. The field K is totally imaginary and $b \geq 1$. Its root discriminant, equal to that of F , is

$$\lambda = 2^{9/4} \sqrt{15015}, \quad \ell = \log \lambda = \frac{9}{4} \log 2 + \frac{1}{2} \log 15015.$$

Write $L_F(s) = \zeta_K(s)/\zeta_F(s)$. We will prove the following bound.

Theorem 4.1. *For all sufficiently large degrees in the constructed field family,*

$$\frac{\log L_F(1)}{d} < 0.042161819.$$

No generalized Riemann hypothesis, effective lower bound for the first such degree, or asymptotic class-number formula is required.

4.1. Finite-Field Comparison and Passage to One. For a number field A , denote its class number, regulator, and number of roots of unity by h_A, R_A, w_A . The analytic class-number formula gives the positive value

$$(4.1) \quad L_F(1) = 2^c \pi^{d-c} \lambda^{-d/2} \frac{h_K R_K w_F}{h_F R_F w_K} > 0.$$

Indeed, the residues of ζ_A are $2^{r_1(A)} (2\pi)^{r_2(A)} h_A R_A / (w_A \sqrt{|\Delta_A|})$, and finite unramifiedness gives $|\Delta_K| = |\Delta_F|^2$. We bound the factor in (4.1). The relative unit and ideal-class normalizations enter later in the geometric argument.

At a prime of F of norm q , quadratic splitting gives respectively

$$\zeta_{K,v}(s) = (1 - q^{-s})^{-2} \quad \text{or} \quad \zeta_{K,v}(s) = (1 - q^{-2s})^{-1}.$$

Both are at most $\zeta_{F,v}(s)^2$ for real $s > 1$. Consequently

$$(4.2) \quad \frac{\log L_F(s)}{d} \leq \frac{\log \zeta_K(s)}{2d}.$$

For a rational prime p with absolute ramification index e and residue degree f in a Galois field, its contribution to the normalized logarithm is

$$(4.3) \quad \Phi_{p,s}(e, f) = \frac{-\log(1 - p^{-fs})}{ef}.$$

Multiplying either e or f by a positive integer cannot increase this expression. Hence, in an extension of finite Galois fields, the normalized zeta function of the larger field is at most that of the smaller field, prime by prime.

Set $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ and $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$. The nontrivial quadratic Hecke character of K/F is unramified at finite places and odd at each real place of F . The Hecke continuation and functional equation theorem [13, 36] gives, up to a positive constant independent of s , the completion

$$(4.4) \quad \mathcal{L}_F(s) = \lambda^{ds/2} \pi^{-bs/2} (2\pi)^{-cs} \Gamma((s+1)/2)^b \Gamma(s)^c L_F(s).$$

It is entire of order one. Taking the quotient of the completed Dedekind zeta functional equations shows that its root number is $+1$.

The zero pairing in the functional equation lets us compare this completion at one with its values to the right of one.

Lemma 4.2. *The function $\mathcal{L}_F(s)$ is nondecreasing on the real interval $[1, \infty)$.*

Proof. The Euler product and functional equation place every zero in $0 \leq \operatorname{Re} s \leq 1$. Center the function at $1/2$ and write $t = s - 1/2$. Its zeros occur in pairs $z, -z$ and in conjugate pairs. Since its order is one, $\sum_{z \neq 0} |z|^{-2} < \infty$. It follows that the paired canonical product and its logarithmic derivative converge locally off the zeros. Evenness makes the remaining linear exponential factor constant. A zero of multiplicity m at the center contributes m/t to the logarithmic derivative. A pair with $z = a + iv$ contributes the real part

$$\operatorname{Re} \frac{2t}{t^2 - z^2} = \frac{2t(t^2 - a^2 + v^2)}{(t^2 - a^2 + v^2)^2 + 4a^2v^2} \geq 0 \quad (t \geq 1/2),$$

because $|a| \leq 1/2$. There are no zeros on $[1, \infty)$: this follows from the Euler product above one and (4.1) at one. The completion is positive there. Its logarithmic derivative is nonnegative, proving the lemma. $\square$

For $\varepsilon > 0$, division of the two completion factors gives

$$(4.5) \quad \frac{\log L_F(1)}{d} \leq \frac{\log \zeta_K(1 + \varepsilon)}{[K : \mathbb{Q}]} + G(\varepsilon, \theta),$$

$$(4.6) \quad G(\varepsilon, \theta) = \frac{\varepsilon}{2}(\ell - \log \pi) - \varepsilon \theta \log 2 + (1 - 2\theta) \log \Gamma(1 + \varepsilon/2) + \theta \log \Gamma(1 + \varepsilon).$$

In particular $G(\varepsilon, \theta) \rightarrow 0$ uniformly for $0 \leq \theta \leq 1/2$ as $\varepsilon \downarrow 0$.

4.2. An Unconditional Asymptotic Prime Budget. The basic inequality for limiting prime counts improves a finite Euler estimate at a fixed abscissa to the following bound at one. Only this inequality is needed, without a Brauer–Siegel equality.

Proposition 4.3. *Suppose that, at $\sigma = 1 + \varepsilon > 1$, every field in the family satisfies*

$$\frac{\log \zeta_K(\sigma)}{[K : \mathbb{Q}]} \leq Y_K(\sigma) \leq Y_*(\sigma).$$

Let e_p^0, f_p^0 be uniform lower bounds for its absolute local indices, and suppose

$$R \geq \sum_p \frac{\log p}{e_p^0(p^{2f_p^0} - 1)}.$$

Then

$$(4.7) \quad \limsup_{d \rightarrow \infty} \frac{\log L_F(1)}{d} \leq Y_*(1 + \varepsilon) + \varepsilon \left(\frac{\ell - \gamma_E - \log(4\pi)}{4} + R \right).$$

Every strictly larger constant is an eventual uniform ceiling.

Proof. Take any sequence with $n = [K : \mathbb{Q}] = 2d \rightarrow \infty$. A diagonal subsequence makes each normalized prime count converge: $\beta_q = \lim N_q(K)/n$, where q ranges over prime powers. The fixed root discriminant makes this an asymptotically exact family. The unconditional Basic Inequality' of [41, Proposition 3.1] is

$$(4.8) \quad \sum_q \beta_q w(q) \leq B, \quad w(q) = 2 \log q \sum_{m \geq 1} \frac{1}{q^m + 1}, \quad B = \frac{\ell - \gamma_E - \log(4\pi)}{2}.$$

To check the normalization, their genus is $g = n\ell/2$, their finite invariants are $\phi_q = 2\beta_q/\ell$, and total imaginary signature gives $\phi_{\mathbb{R}} = 0$, $\phi_{\mathbb{C}} = 1/\ell$. The proposition is unconditional, with archimedean constant $\gamma_E + \log(4\pi)$.

Put $a_s(q) = -\log(1 - q^{-s})$ and $Z_{\infty}(s) = \sum_q \beta_q a_s(q)$. The series converges at one, since $a_1(q)/w(q) \leq (q+1)/(2(q-1)\log q)$ is bounded for $q \geq 2$. For every fixed $\tau > 1$, dominated convergence in the normalized Euler products gives

$$\lim_n \frac{\log \zeta_K(\tau)}{n} = Z_{\infty}(\tau).$$

One can dominate prime by prime by $-\log(1 - p^{-\tau})$. Equation (4.5), followed by $\tau \downarrow 1$, then gives $\limsup d^{-1} \log L_F(1) \leq Z_{\infty}(1)$. The uniform vanishing of G and monotone convergence justify this order of limits. The finite-field Euler products are used only to the right of one.

For every prime power q ,

$$\begin{aligned} a_1(q) - a_{1+\varepsilon}(q) &\leq \frac{\varepsilon \log q}{q-1}, \\ \frac{\log q}{q-1} - \frac{w(q)}{2} &= \log q \sum_{m \geq 1} \frac{1}{q^m(q^m + 1)} \leq \frac{\log q}{q^2 - 1}. \end{aligned}$$

For a Galois field, the normalized contribution above p to the last weighted sum is $\log p/[e_p(p^{2f_p} - 1)]$. The local floors and Fatou's lemma bound its limiting sum by R . Combining this with (4.8) proves $Z_{\infty}(1) \leq Z_{\infty}(1 + \varepsilon) + \varepsilon(B/2 + R)$. This proves the assertion on every such subsequence. If a strictly larger constant were not an eventual uniform bound, a sequence of violating fields would have a diagonal subsequence of the kind just considered, giving a contradiction. $\square$

4.3. The Complete Finite-Field Euler Factorization. Let N be the degree-16384 retained field of Proposition 3.2, and set $S' = \{2, 3, 5, 7, 11, 13, 17\}$. A superscript S' removes these Euler factors. Proposition 3.4 gives all irreducible representations of $\text{Gal}(N/\mathbb{Q})$ and realizes each nonlinear factor as an induced nontrivial finite-order Hecke character. There are 128 linear, 256 quadratic, 456 quartic, and 124 octic factors. Among the quartics, eight have pure gamma parity, and every other nonlinear factor is mixed. The regular representation identity is

$$(4.9) \quad \zeta_N(s) = \prod_{\rho} L(s, \rho)^{\dim \rho}, \quad 128 + 256 \cdot 4 + 456 \cdot 16 + 124 \cdot 64 = 16384.$$

For real $s > 1$ the factors are nonzero. Real factors are positive: their Euler products tend to one as $s \rightarrow \infty$ and cannot cross zero. Conjugate factors have product $|L(s, \rho)|^2$. Thus

$$(4.10) \quad Y(s) := \frac{\log \zeta_N^{S'}(s)}{16384} = \frac{1}{16384} \sum_{\rho} \dim \rho \log |L^{S'}(s, \rho)|.$$

Each conjugate pair is counted twice in this sum, with the same representation degree. An upper bound for each modulus may be substituted after this identity, even when some logarithms are negative.

The exact arithmetic input for one factor consists of its conductor Q , gamma parities, root number when proved to be one, an inducing radicand, genus twist, and the exceptional denominators

$$D_{\rho,p}(T) = \det(1 - T\rho(\text{Frob}_p) \mid V_{\rho}^{I_p}).$$

Their construction, including the cancellation of quartic twists at 5 and 13, is given in Proposition 3.4. The analytic evaluations below use the representations constructed there, including their conductors and local characters. Formal compatibility among a set of Euler polynomials would not identify these functions.

4.4. Exact Coefficients and Their Enumeration. Write $L(s, \rho) = \sum_{n \geq 1} a_n n^{-s}$ and $r = \dim \rho$. At every prime, the reciprocal roots of $D_{\rho,p}$ have modulus one, and there are at most r of them. Hence

$$(4.11) \quad |a_n| \leq d_r(n), \quad \sum_{n \geq 1} d_r(n) n^{-\beta} = \zeta(\beta)^r \quad (\beta > 1).$$

The finite-image representation also makes every coefficient used here an exact Gaussian integer.

To construct the coefficients outside S' , let $v_p \in \mathbb{F}_2^7$ be genus Frobenius, and let B_q be the polar form of the corresponding central character. Outside its radical, Frobenius has eigenvalues in opposite pairs and square κI , where $\kappa \in \{1, -1\}$. Its denominator is $(1 - \kappa T^2)^{r/2}$, before the explicitly supplied quartic twist. The coefficients at exponents $2j$ are $\binom{j+r/2-1}{r/2-1} \kappa^j$, and odd exponents vanish. Inside the radical, Frobenius is scalar, and its denominator is $(1 - \omega T)^r$, where $\omega \in \{1, -1, i, -i\}$ is determined by the inducing field and twist. Its coefficients are $\binom{j+r-1}{r-1} \omega^j$.

At a scalar prime the inducing multiquadratic base splits completely. For the positive quadratic induction, the scalar sign is obtained by evaluating the radicand at these embeddings modulo p and taking quadratic residue symbols. These symbols agree because the represented Frobenius is scalar. There is an exact procedure even at a prime dividing the norm of the displayed radicand. For an integral radicand η , let $m = v_p(N\eta)$. Each embedding valuation is between zero and m . Lift the base square roots to modulus p^{m+1} by Hensel's lemma. This determines every embedding valuation and normalized nonzero unit residue. The valuations are even, by unramifiedness outside S' . The normalized quadratic symbols give the required signs. Rational square normalization does not change the extension. This procedure treats every norm prime, rather than assuming that a finite list of ordinary exceptions is exhaustive.

The genus and twist characters are evaluated exactly by Kronecker symbols and by discrete logarithms in $(\mathbb{Z}/5\mathbb{Z})^\times$ or $(\mathbb{Z}/13\mathbb{Z})^\times$, with $\xi_p(2) = i$. The genus characters have common period 120120, so lookup tables may use this period or any multiple of it. At $p \in S'$, the verified polynomial $D_{\rho,p}$ supplies the recurrence

$$a_{pj} = - \sum_{k=1}^{\min(j,r)} [T^k] D_{\rho,p}(T) a_{pj-k}, \quad a_1 = 1.$$

Multiplicativity then determines all coefficients.

To enumerate the nonzero coefficients through the cutoff N_0 , first list all primes up to N_0 by a segmented sieve using every sieving prime through $\lfloor \sqrt{N_0} \rfloor$. Assign an ordinary nonscalar prime the first possible support location p^2 , a scalar prime the location p , and an exceptional prime the safe lower bound p . Sort primes by these locations. Recursion appends only later primes in that order, considering every admissible prime power and continuing across intermediate zero coefficients. Unique factorization then visits each supported integer exactly once. If a first support location

exceeds the remaining quotient, that prime and every later prime can be skipped. These observations prove that the enumeration is complete.

Signed and absolute moments use arbitrary-precision integers. Fixed-width coefficient arithmetic is checked before conversion, and intermediate products are formed in a wider type. An overflow aborts the calculation. Modular multiplication uses a wide product or a checked Montgomery reduction as appropriate. The coefficients, support counts, and moments are computed exactly.

4.5. The Linear Factors. The 128 linear factors are the primitive quadratic Dirichlet functions of the genus field $E = \mathbb{Q}(i, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{11}, \sqrt{13})$, including the principal factor $\zeta(s)$. For a signed squarefree product D_0 of these radicands, its fundamental discriminant is $D = D_0$ if $D_0 \equiv 1 \pmod{4}$, and $D = 4D_0$ otherwise. The character is the Kronecker symbol $(D/\cdot)$. The implementation checks its primitive conductor and its values on explicit generators of every prime-power component of the unit group. These values identify the character independently of its library index.

The finite identity

$$(4.12) \quad L(s, \chi) = q^{-s} \sum_{a=1}^q \chi(a) \zeta(s, a/q)$$

reduces evaluation to Hurwitz zeta functions. Euler–Maclaurin summation gives the explicit enclosure

$$\begin{aligned} \zeta(s, x) &= \sum_{k=0}^{M-1} (k+x)^{-s} + \frac{(M+x)^{1-s}}{s-1} + \frac{1}{2}(M+x)^{-s} \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{(2j)!} (s)_{2j-1} (M+x)^{1-s-2j} + \mathcal{R}, \\ |\mathcal{R}| &\leq \frac{2\zeta(2J)(s)_{2J}}{(2\pi)^{2J}(s+2J-1)} (M+x)^{1-s-2J}, \quad s > 1, x > 0. \end{aligned}$$

The remainder follows from the periodic Bernoulli bound $|\tilde{B}_{2J}(t)| \leq 2(2J)! \zeta(2J)/(2\pi)^{2J}$. For rigorous high-precision evaluation by Euler–Maclaurin summation, see [15]. The supplied evaluation uses Arb/FLINT ball arithmetic [16], retaining the endpoints of each real value. Removing S' multiplies a factor by $\prod_{p \in S'} (1 - \chi(p)p^{-s})$.

4.6. An Approximate Functional Equation for Mixed Factors. Numerical evaluation from gamma factors and a functional equation is developed in [4]. We use this approach with explicit kernel and tail bounds for the factors constructed above. For a mixed factor of degree $2m$, where $m = 1, 2, 4$, its completion is

$$\Lambda(s) = Q^{s/2} \Gamma_{\mathbb{R}}(s)^m \Gamma_{\mathbb{R}}(s+1)^m L(s) = 2^m Q^{s/2} (2\pi)^{-ms} \Gamma(s)^m L(s).$$

It is entire and satisfies $\Lambda(s) = \varepsilon_{\rho} \overline{\Lambda(1-\bar{s})}$, with $|\varepsilon_{\rho}| = 1$. For a real quadratic induction the Dedekind zeta quotient gives $\varepsilon_{\rho} = 1$.

Let $k_m(x)$ have Mellin transform $\Gamma(z)^m$, and put

$$H_{m,a}(x) = \int_1^{\infty} u^{a-1} k_m(xu) du, \quad t = \frac{(2\pi)^m}{\sqrt{Q}}, \quad F_m = \frac{t^s}{\Gamma(s)^m}.$$

These kernels are positive. More explicitly,

$$(4.13) \quad k_m(x) = \int_{(0,\infty)^{m-1}} \exp\left(-u_1 - \cdots - u_{m-1} - \frac{x}{u_1 \cdots u_{m-1}}\right) \frac{du_1 \cdots du_{m-1}}{u_1 \cdots u_{m-1}},$$

with $k_1(x) = e^{-x}$. Taking the Mellin transform under this positive integral proves the asserted transform and also complete monotonicity in x . In particular $k_2(x) = 2K_0(2\sqrt{x})$ [27, 10.32.10, 10.43.19].

Splitting the Mellin integral at one gives a common approximate functional equation for the mixed factors.

Lemma 4.4. *For real $s > 1$ and every nonlinear mixed factor,*

$$(4.14) \quad L(s) = F_m \sum_{n \geq 1} a_n H_{m,s}(tn) + \varepsilon_\rho F_m \sum_{n \geq 1} \bar{a}_n H_{m,1-s}(tn).$$

Proof. The function $\Theta(u) = \sum a_n k_m(tnu)$ has Mellin transform $\Gamma(z)^m t^{-z} L(z) = 2^{-m} \Lambda(z)$. Mellin inversion and the completed functional equation give $\Theta(u) = \varepsilon_\rho u^{-1} \overline{\Theta(1/u)}$. The contour shift has no residues because the completion is entire, and the gamma factors give the needed vertical decay. Splitting the Mellin integral at one and substituting $u \mapsto 1/u$ below one yields (4.14). The positive integral representation and (4.11) justify the sums in either exponentially decaying half. $\square$

For degree two this uses $H_{1,a}(x) = \int_1^\infty u^{a-1} e^{-xu} du$. For the mixed quartic implementation it is convenient to write

$$I_b(y) = \int_1^\infty u^b K_0(yu) du, \quad H_{2,a}(x) = 4I_{2a-1}(2\sqrt{x}).$$

Its prefactor is $4t^s/\Gamma(s)^2$ when the stored kernel is $I_{2s-1} + I_{1-2s}$. This identity accounts for the factor four in the quartic evaluation. For octics the stored kernel is $H_{4,a}$ itself, with prefactor $t^s/\Gamma(s)^4$.

4.7. Tail Bounds and Absolute Coefficient Prefixes. For $\beta \geq \max(s, 1-s)$, $H_{m,a}(x) \leq H_{m,\beta}(x)$ when $a \leq \beta$, and

$$x^\beta H_{m,\beta}(x) = \int_x^\infty v^{\beta-1} k_m(v) dv$$

is decreasing. If $M_{N_0}(\beta)$ bounds $\sum_{n > N_0} |a_n| n^{-\beta}$, either omitted half of (4.14) is at most

$$(4.15) \quad F_m(N_0 + 1)^\beta H_{m,\beta}(t(N_0 + 1)) M_{N_0}(\beta).$$

One may always use $M_{N_0}(\beta) = \zeta(\beta)^{2m}$.

The quadratic and quartic evaluations also use elementary envelopes. For the exponents in question, with $1 < s < 1.01$,

$$(4.16) \quad H_{1,s}(x), H_{1,1-s}(x) \leq V(x) := e^{-x}(1 + 1/x)/x,$$

$$(4.17) \quad I_{2s-1}(y), I_{1-2s}(y) \leq U(y) := \sqrt{\pi/2} e^{-y}(1 + 1/y)y^{-3/2}.$$

The first follows by replacing u^{s-1} and u^{-s} by u . For the second use $K_0(yu) \leq \sqrt{\pi/(2yu)} e^{-yu}$ and $u^{b-1/2} \leq u$ for $b \leq 3/2$. The functions $n^\beta V(tn)$ and $n^\beta U(2\sqrt{tn})$ decrease respectively when $tn > \beta - 1$ and $2\sqrt{tn} > 2\beta - 3/2$. Thus substituting these envelopes and the divisor majorant gives valid tails even when the chosen β exceeds an exponent covered by the envelope. The quartic prefactor includes the factor four from the conversion between the kernels. When we sum both halves, we include both tail allowances.

For the octics a sharper absolute series is available. Let $W = \text{rad } B_q$. Then $[E^W : \mathbb{Q}] = 64$. At an ordinary prime, absolute coefficients have local series $(1 - T)^{-8}$ if $v_p \in W$, and $(1 - T^2)^{-4}$ otherwise. These are precisely the local factors of $\zeta_{E^W}^{S'}(\beta)^{1/8}$. For every exceptional denominator in the supplied arithmetic table, an exact Gaussian-polynomial comparison supplies nonnegative integers u_p, v_p and a fourth root of unity ω_p such that

$$(4.18) \quad D_{\rho,p}(T) = (1 - \omega_p T)^{u_p} (1 - \omega_p^2 T^2)^{v_p}.$$

The check enumerates $u_p, v_p \geq 0$ with $u_p + 2v_p \leq 8$ and the four possible ω_p , comparing all polynomial coefficients exactly. After the rotation $T \mapsto \omega_p^{-1}T$, the reciprocal has nonnegative coefficients. Consequently the full absolute series is exactly

$$(4.19) \quad M_\rho(\beta) := \sum_{n \geq 1} |a_n| n^{-\beta} = \zeta_{EW}^{S'}(\beta)^{1/8} \prod_{p \in S'} (1 - p^{-\beta})^{-u_p} (1 - p^{-2\beta})^{-v_p}.$$

The same rotations show that each nonzero coefficient lies on a Gaussian coordinate axis, so $|a_n| = |\operatorname{Re} a_n| + |\operatorname{Im} a_n|$ exactly. The genus subfield zeta value is evaluated using its 64 quadratic characters by (4.12).

We subtract a rigorous lower bound for the finite absolute prefix. On a bin with integer midpoint m_0 and radius r_0 , record exact moments $A_j = \sum |a_n| (n - m_0)^j$, $0 \leq j \leq J = 20$. Let $h = \lceil \beta \rceil$ and $\rho_0 = r_0/m_0 < 1/16$. The degree- J binomial expansion of $n^{-\beta}$ has error at most

$$(4.20) \quad A_0 m_0^{-\beta} \binom{J+h}{h-1} \frac{\rho_0^{J+1}}{(1-\rho_0)^h}.$$

Indeed $(\beta)_j/j! \leq \binom{j+h-1}{h-1}$, and after the first omitted coefficient their ratios are bounded by the coefficients of $(1-\rho_0)^{-h}$. Sum the interval polynomials and subtract these errors to obtain $P_{N_0}(\beta) \leq \sum_{n \leq N_0} |a_n| n^{-\beta}$. Then $M_\rho(\beta) - P_{N_0}(\beta)$ is an admissible $M_{N_0}(\beta)$ in (4.15). The octic evaluation selects the smallest valid enclosure among $\beta = 5/4, 3/2, 2, 5/2, 3$. Each candidate is an upper bound, so this selection preserves the inequality.

4.8. The Eight Pure Quartic Factors. The pure factors in Proposition 3.4 are real quadratic Hecke inductions with root number one, conductor $Q = 240240^2$, and gamma vector (a, a, a, a) , $a \in \{0, 1\}$, four of each parity. Their completion is $Q^{s/2} \Gamma_{\mathbb{R}}(s+a)^4 L(s)$. Put

$$t = \pi^2/\sqrt{Q}, \quad b_0 = (s+a)/2, \quad b_1 = (1-s+a)/2, \quad F = \frac{t^{s+a}}{\Gamma(b_0)^4}, \quad K_b = H_{4,b}.$$

Then the exact functional equation is

$$(4.21) \quad L(s) = F \sum_{n \geq 1} a_n n^a (K_{b_0}(t^2 n^2) + K_{b_1}(t^2 n^2)).$$

To verify the normalization, the Mellin transform of $\sum a_n n^a k_4(t^2 n^2 u)$ is

$$\Gamma(z)^4 t^{-2z} L(2z-a) = Q^{a/2} \Lambda(2z-a).$$

Reflection is about $z = (a+1/2)/2$. Splitting at one gives (4.21), with no extra factor two. For $\beta \geq \max(s, 1-s)$, put $B = (\beta+a)/2$. Either omitted side is bounded by

$$(4.22) \quad F(N_0+1)^{\beta+a} K_B(t^2(N_0+1)^2) \zeta(\beta)^4.$$

This follows from the same decreasing-integral argument as (4.15). The exact cutoff here is $N_0 = 8\sqrt{Q} = 1921920$.

4.9. Kernel and Moment Evaluation. We evaluate the kernels on finite grids and bound the interpolation errors. For the four-gamma kernel, let $H_n^{(j)} = \sum_{k=1}^n k^{-j}$ and

$$L_n = 4(H_n^{(1)} - \gamma_E) - \log x, \quad P_{3,n} = L_n^3/6 + 2L_n(\zeta(2) + H_n^{(2)}) + 4(H_n^{(3)} - \zeta(3))/3, \\ P_{2,n} = L_n^2/2 + 2(\zeta(2) + H_n^{(2)}).$$

Residues at the order-four poles of $\Gamma(z)^4$ give the convergent series

$$(4.23) \quad k_4(x) = \sum_{n \geq 0} \frac{x^n}{(n!)^4} P_{3,n},$$

$$(4.24) \quad K_a(x) = \Gamma(a)^4 x^{-a} - \sum_{n \geq 0} \frac{x^n}{(n!)^4} \left(\frac{P_{3,n}}{n+a} + \frac{P_{2,n}}{(n+a)^2} + \frac{L_n}{(n+a)^3} + \frac{1}{(n+a)^4} \right).$$

The second identity initially follows for $a > 0$ and extends to $-1/100 < a < 0$ by analytic continuation. None of the exponents evaluated here is zero or a negative integer, so no colliding-pole formula is needed.

To bound the tails of these series, for $n \geq 2$ set

$$B_n = |\log x| + 4(1 + \log n + \gamma_E), \quad U_{3,n} = B_n^3/6 + 4\zeta(2)B_n + 4\zeta(3)/3, \quad U_{2,n} = B_n^2/2 + 4\zeta(2).$$

They bound $|L_n|, |P_{3,n}|, |P_{2,n}|$ in the corresponding order. Also $B_{n+1} \leq (1 + 1/n)B_n$, so

$$r_n = \frac{x}{(n+1)^4} (1 + 1/n)^3$$

is an upper bound for the subsequent term ratios of the majorants. Once $r_n < 1$, the entire omitted tail of (4.23) is at most

$$\frac{x^n}{(n!)^4} \frac{U_{3,n}}{1 - r_n}.$$

For (4.24), provided $a > -1/100$, it is at most

$$\frac{x^n}{(n!)^4} \frac{1}{1 - r_n} \left(\frac{U_{3,n}}{n-1} + \frac{U_{2,n}}{(n-1)^2} + \frac{B_n}{(n-1)^3} + \frac{1}{(n-1)^4} \right).$$

All these expressions are evaluated outward. The first four Taylor coefficients of k_4 are obtained by the same argument on $|z - x| \leq x/2$, replacing x by $3x/2$ in the numerator bound and $|\log x|$ by $|\log x| + \log 2 + \pi/2$. Cauchy's estimate bounds the error in the j^{th} coefficient by the uniform tail times $(2/x)^j$.

Higher Taylor coefficients follow from

$$(4.25) \quad (x\partial_x)^4 k_4 = xk_4, \quad xK'_a + aK_a = -k_4.$$

These identities follow directly from the Mellin transform and the defining integral. They give finite recurrences using rational operations and the enclosed initial values. The two-gamma kernel uses $K(x) = K_0(2\sqrt{x})$, $J_b(x) = I_b(2\sqrt{x})$, with

$$xK'' + K' - K = 0, \quad 2xJ'_b + (b+1)J_b = -K.$$

If k_j, j_j denote Taylor coefficients at x_0 , the recurrences are

$$k_{j+2} = \frac{k_j - (j+1)^2 k_{j+1}}{x_0(j+1)(j+2)}, \quad j_{j+1} = \frac{-k_j - (2j+b+1)j_j}{2x_0(j+1)}.$$

The initial values are $K_0(2\sqrt{x_0})$ and $-K_1(2\sqrt{x_0})/\sqrt{x_0}$. The one-gamma case uses the elementary exponential and $xH'_{1,a} + aH_{1,a} = -e^{-x}$.

The grids have consecutive endpoints $x_0, 31x_0/32$. For any positive Laplace kernel H in use, its holomorphic extension obeys $|H(z)| \leq H(x_0/2)$ on $|z - x_0| \leq x_0/2$. Hence a degree- J Taylor polynomial has error at most

$$(4.26) \quad H(x_0/2) \frac{16^{-J-1}}{1 - 1/16}$$

throughout the bin. The backward Taylor remainder is nonnegative by complete monotonicity, allowing propagation of the exponential and Bessel kernels from a certified large-argument interval. Their starting values are bounded by (4.16) and (4.17). The four-gamma kernel instead uses the

residue series at each center. Grid endpoints, their integer floors, and complete coverage from 1 to the exact coefficient cutoff are checked with outward intervals. The calculation is rejected if an integer floor is unresolved.

For a linear kernel argument $x = tn$, translating its degree-20 Taylor polynomial at x_0 to the integer midpoint m_0 gives

$$[(n - m_0)^j] P(tn) = t^j \sum_{h=j}^{20} [(x - x_0)^h] P(x) \binom{h}{j} (tm_0 - x_0)^{h-j}.$$

Contract this polynomial against the exact real and imaginary moments. Multiplying (4.26) by the sum of absolute coefficients in the bin bounds its contribution to the error. For the pure quartics the substitution is quadratic:

$$t^2 n^2 - x_0 = (t^2 m_0^2 - x_0) + 2t^2 m_0(n - m_0) + t^2(n - m_0)^2.$$

A degree-20 kernel polynomial thus needs moments through degree 40. These moments and their absolute masses include n^a exactly once. The uniform remainder still has degree 20 in the kernel variable, so (4.26) applies before composition.

Suppose A, B are the finite primary and conjugated dual sums, and E_A, E_B bound their interpolation and infinite-tail errors. A known root number one gives $|L(s)| \leq |A + B| + E_A + E_B$. With an unknown phase one always has $|L(s)| \leq |A| + |B| + E_A + E_B$. Every modulus, gamma value, logarithm, and conversion to an endpoint retains its rounding enclosure.

4.10. Recovering the Mixed Quadratic and Quartic Phase. For the complex quadratic and mixed quartic factors, a second Mellin cutoff can sharpen this bound by enclosing the root number. Split the Mellin integral at $y > 0$ instead of one. The two exact terms are

$$A_y = F_m y^s \sum a_n H_{m,s}(tyn), \quad B_y = F_m y^{s-1} \sum \bar{a}_n H_{m,1-s}(tn/y), \quad L(s) = A_y + \varepsilon_\rho B_y.$$

Take $y = (31/32)^{22}$, so shifted kernels reuse the same exact moment bins. Any terminal part outside the stored kernel range is put into the tail with its actual first omitted integer.

For finite approximations, write $U = A_y - A_1$, $V = B_1 - B_y$ and let η, Δ bound their respective errors. If $|V| > \Delta$, then

$$(4.27) \quad \varphi = U/V, \quad |\varepsilon_\rho - \varphi| \leq r_\varphi := \frac{\eta + |\varphi|\Delta}{|V| - \Delta}.$$

This follows by writing the exact quotient as $(U + u)/(V + v)$, with $|u| \leq \eta$, $|v| \leq \Delta$. If each balanced side has error at most E_1 , the resulting bound is

$$(4.28) \quad |L(s)| \leq |A_1 + \varphi B_1| + 2E_1 + |B_1| r_\varphi.$$

The coefficient $2E_1$ uses the known identity $|\varepsilon_\rho| = 1$, not an assumption that $|\varphi| = 1$. All distances from zero and radii are enclosed outward. If the denominator cannot be separated from zero, the phase-free bound is retained. Complex octic factors use the phase-free bound directly.

4.11. The Certified Factor Sums. The arithmetic table and exact moment files in the supplementary certificate determine every factor just described. The abscissa is the exact rational

$$\sigma = \frac{12001}{12000}.$$

For each row, the evaluator applies the appropriate AFE, includes interpolation and both infinite tails, and removes the exceptional factors by

$$\log |L^{S'}(\sigma, \rho)| \leq \log U_\rho + \sum_{p \in S'} \log |D_{\rho,p}(p^{-\sigma})|,$$

where U_ρ is its proved modulus upper bound. The octic cutoffs are $N_0 = \min(3 \cdot 10^9, \lfloor \sqrt{Q}/40 \rfloor)$. The exact integers for the quadratic and mixed quartic rows are specified in their moment records and individually checked when constructing the bins and tails. The pure quartic cutoff was given in Section 4.8.

The following table gives rational upper bounds for the sums of logarithms. The number of factors includes both members of each conjugate pair. An interval enclosing an AFE upper bound encloses that bound, and need not give a two-sided enclosure of comparable width for the L -value itself.

Type	Number	Upper bound for $\sum \log L^{S'}(\sigma, \rho) $
Linear	128	7.337578130846530221
Mixed quadratic	256	0.400886873183882259
Mixed quartic	448	-0.046453348154273950
Pure quartic	8	0.058145052434373409
Mixed octic	124	-0.111176963043772737

Exact addition of their dimension-weighted upper endpoints in (4.10) gives

$$(4.29) \quad Y(\sigma) \leq 0.00044535540710354679437255859375 < 0.000445355407103547.$$

The coefficient algorithms, kernel bounds, and tail estimates above specify how to verify these inequalities. The supplementary program checks the exact row identities and partitions before performing the interval evaluations. Its moment mode regenerates the newest 64 sector moment files from the coefficient algorithms. Earlier moment records remain finite inputs, and the data manifest checks their identities.

4.12. Finite Corrections beyond the Retained Field. To compare N with the whole tower, restore S' using the actual local indices of K . Its exact contribution is

$$(4.30) \quad B_D(\sigma) = -\frac{\log(1-2^{-4\sigma})}{32} - \sum_{p=3,5} \frac{\log(1-p^{-2\sigma})}{4}$$

$$(4.31) \quad - \sum_{p=7,11,13} \frac{\log(1-p^{-4\sigma})}{8} - \frac{\log(1-17^{-4\sigma})}{4}.$$

This restores the deleted factors using the local indices of K . The indices of N at S' may be different.

First consider $p \leq 10^4$, $p \notin S'$, with nonzero genus vector v_p . Its square is detected in the full retained degree-two quotient if some form in the twelve-dimensional central space has $q(v_p) = 1$. If all seven completed forms instead have $q(v_p) = 0$, then $f_N(p) = 2$ whereas $4 \mid f_K(p)$. Define $\mathcal{P}_4$ by precisely these conditions. It contains 45 primes. Equation (4.3) gives the saving

$$(4.32) \quad S_4 = \frac{1}{4} \sum_{p \in \mathcal{P}_4} \log \frac{1+p^{-2\sigma}}{1-p^{-2\sigma}} > 0.00003724741189544274807446.$$

Primes for which the completed field already has order four are excluded from this sum.

The second correction uses primes that split in E and in N , but not in the full twelve-dimensional retained central field. Here $f_N(p) = 1$ and $f_K(p)$ is even, so the saving is

$$(4.33) \quad S_{\text{cen}} = \sum_{\substack{p \leq 10^{12} \\ p \text{ split in } N, p \text{ nonsplit in the full field}}} \operatorname{atanh}(p^{-\sigma}).$$

The support has zero genus vector, so it is disjoint from $\mathcal{P}_4$. Both supports exclude S' . No saving from an uncomputed tail is claimed.

To compute (4.33), enumerate the genus-split primes. Their residues are the 180 classes modulo 120120 satisfying $p \equiv 1 \pmod{8}$ and being nonzero squares modulo each of 3, 5, 7, 11, 13. Construct them by the Chinese remainder theorem. For each class, sieve the numbers $120120k + r$ through $X = 10^{12}$ by all primes up to $\sqrt{X}$, beginning at the square of each sieving prime. Every surviving number above 13 is prime, since a composite would have a prime divisor at most its square root. The classes are disjoint and none contains a prime dividing the modulus.

For a surviving prime, evaluate the 17 norm radicands of Section 3.1 using exact modular square roots. The symbol of $x + y\sqrt{A}$ is independent of the choice of root: its product with the conjugate is Bz^2 , a nonzero square at a genus-split prime. All catalog B are supported on 2, 3, 5, 7, 11, 13 and all catalog z are 1 or 2. Thus no prime in this census divides Bz , and no zero residue or unhandled norm prime occurs in these raw predicates. The seven completed and twelve full-field product predicates in Proposition 3.2 are then exact splitting tests. Order the completed tests first and count only the first nonzero predicate. A hit in the first seven tests is already accounted for by N . A later hit belongs to (4.33). A prime on which every predicate vanishes is not counted. This proves that the computed complement contains each eligible prime exactly once.

For outward summation, use disjoint bins $l < p \leq h$, beginning at $l = 13$ and with next endpoint $h = \min(X, \max(l + 1, \lceil 1001l/1000 \rceil))$. For complement primes in a bin record their exact number n and $u = \sum \lfloor 10^{30}/p \rfloor$. With $\varepsilon = \sigma - 1$, their contribution is enclosed between

$$(4.34) \quad h^{-\varepsilon} \frac{u}{10^{30}} \quad \text{and} \quad l^{-\varepsilon} \frac{u + n}{10^{30}} \frac{1}{1 - l^{-2}}.$$

Indeed $x \leq \operatorname{atanh} x \leq x/(1 - x^2)$ for $0 < x < 1$, and $\sigma > 1$. The only rounded quantities in the producer are the explicit floors, whose full error is included in $u + n$. All counts and reciprocal numerators use exact integers. For the fixed-width census implementation, counts are at most X , and the total reciprocal numerator is less than $10^{30}(1 + \log X) < 2^{105}$. Thus its 64-bit counts and 128-bit accumulators cannot overflow at this cutoff. Modular products use the wider reduction described above.

The full sieve finds 293772662 genus-split primes. Of these, 293705608 are detected by the full central field, and 291485548 are already detected by N . The complement contains 2220060 primes. Summing (4.34) outward gives

$$(4.35) \quad 0.00004201663320373563276702 < S_{\text{cen}} < 0.00004201663670363397482805.$$

An independent prime-list and modular-arithmetic calculation through 10^7 checks every bin, first-hit count, and reciprocal numerator. Its complement has 31 primes. This finite comparison checks the implementation but does not replace the full sieve or its completeness argument.

Primewise comparison now proves, for every retained tower field,

$$(4.36) \quad \frac{\log \zeta_K(\sigma)}{[K : \mathbb{Q}]} \leq Y(\sigma) + B_D(\sigma) - S_4 - S_{\text{cen}}.$$

Remark 4.5 (Fixed-field form). The same primewise comparison applies to the retained field M of Proposition 3.2 itself, of degree 2^{19} , and gives

$$\frac{\log \zeta_M(\sigma)}{2^{19}} \leq Y(\sigma) + B_D(\sigma) - S_4 - S_{\text{cen}}.$$

Since M contains N , equation (4.3) bounds its normalized zeta logarithm prime by prime by that of N outside S' . At S' we restore the factors in B_D using the indices of M . These are $(e, f) = (8, 4)$ at 2, $(2, 2)$ at 3 and 5, $(2, 4)$ at 7, 11, 13, and $f = 4$ at 17, 19, 23, 29, 31. To justify these indices, Proposition 3.2 identifies M with the field of G/D_3G , which retains every local cut of Section 2. A direct local computation from the radicands also confirms them. They give the terms in (4.31) and the local floors used below.

The conditions defining $\mathcal{P}_4$ and the census complement are the tests (3.13) on the central quadratic forms and the twelve full-field predicates. These are tests in M . Hence a prime in $\mathcal{P}_4$ has $f_M(p) = 4$, and a complement prime has $f_M(p) = 2$. Both savings S_4 and S_{cen} are thus realized in M . Each term in the following absolutely convergent series decreases in $e_M(p)$ and $f_M(p)$, so the same local floors give

$$-\frac{\zeta'_M(2)}{\zeta_M} \cdot \frac{1}{2^{19}} = \sum_p \frac{\log p}{e_M(p)(p^{2f_M(p)} - 1)} \leq R$$

with R as in (4.37). Consequently the fixed-field quantity

$$H_M := \frac{\log \zeta_M(\sigma)}{2^{19}} + (\sigma - 1) \left(\frac{\ell - \gamma_E - \log(4\pi)}{4} - \frac{\zeta'_M(2)}{\zeta_M} \cdot \frac{1}{2^{19}} \right)$$

satisfies, with L as in (4.39) and the five allowances (4.29), (4.38), (4.32), (4.35) and (4.40),

$$H_M \leq Y(\sigma) + B_D(\sigma) - S_4 - S_{\text{cen}} + (\sigma - 1)L < 0.04216181893948094953138458.$$

The accompanying Lean formalization assumes this inequality for H_M with the relaxed threshold 0.042165819. Under that hypothesis, it proves the transfer from M to the tower fields and Proposition 4.3. The numerical bound for H_M itself is not proved in the formalization.

4.13. The Numerical Prime-Budget Bound. We specify the local floors used in Proposition 4.3. For $p \leq 10^4$, take $e_2^0 = 8$, $e_p^0 = 2$ at $p = 3, 5, 7, 11, 13$, and $e_p^0 = 1$ elsewhere. Use the exact retained residue degrees 4 at 2, 2 at 3, 5, and 4 at 7, 11, 13, 17, 19, 23, 29, 31. At any other such prime use $f_p^0 = 4$ if its square is nonzero in the full central quotient, $f_p^0 = 2$ if its genus vector is nonzero, and $f_p^0 = 1$ otherwise. We apply the square test only when the genus vector is nonzero. For a genus-split prime with a nontrivial central image, we keep the weaker floor one. All these are lower bounds supplied by the actual finite quotients in Proposition 3.2. Above 10^4 use both floors one.

Since $\log x/(x^2 - 1)$ decreases on $[2, \infty)$, the integer tail is bounded by its integral. It is enough to take

$$(4.37) \quad R = \sum_{p \leq 10^4} \frac{\log p}{e_p^0(p^{2f_p^0} - 1)} + \frac{\log 10^4 + 1}{10^4(1 - 10^{-8})}.$$

All finite floors are reconstructed by exact genus and square tests. Outward evaluation gives

$$(4.38) \quad B_D(\sigma) < 0.041727023150898786,$$

$$(4.39) \quad L := \frac{\ell - \gamma_E - \log(4\pi)}{4} + R < 0.824453118933538946712643997041,$$

$$(4.40) \quad (\sigma - 1)L < 0.00006870442657779491222606.$$

The slope is independent of the abscissa, and the debit uses exactly $\sigma - 1 = 1/12000$.

Combining Proposition 4.3 with (4.36), and then using the five conservative rational bounds (4.29), (4.38), (4.32), (4.35), and (4.40), gives

$$\begin{aligned} \limsup_{d \rightarrow \infty} \frac{\log L_F(1)}{d} &\leq Y(\sigma) + B_D(\sigma) - S_4 - S_{\text{cen}} + (\sigma - 1)L \\ &< 0.04216181893948094953138458 \\ &< 0.042161819. \end{aligned}$$

The middle number is the exact sum of the five rational bounds, with the saving terms subtracted using their lower endpoints. The eventual uniformity in Proposition 4.3 now proves Theorem 4.1.

5. NORM-ONE MASS AND THE GEOMETRIC TRANSFER

Let K/F be quadratic with involution ι , let K be totally imaginary, and let F have signature (b, c) , where $b > 0$. We keep these assumptions in Sections 6 and 7. Write

$$d = [F : \mathbb{Q}] = b + 2c, \quad \theta = c/d, \quad |\Delta_K| = \lambda^{2d}, \quad |\Delta_F| = \lambda^d, \quad \ell = \log \lambda.$$

Assume that K/F is unramified at all finite places. The discriminants are field discriminants, and archimedean integrals use Lebesgue measure on unweighted complex coordinates. We count ordered edges and divide by two to obtain the stated bounds for unordered edges.

We use Dirichlet's unit theorem, Kronecker's theorem, trace duality, Hilbert's theorem 90, and the analytic class-number formula [23, 24].

5.1. The Unit Mass. Put $E_K = \mathcal{O}_K^\times$, $E_F = \mathcal{O}_F^\times$, $U = \ker(N : E_K \rightarrow E_F)$, and $I_U = [E_F : N(E_K)]$. At each complex place of F , choose its two extensions to K and set

$$l_j(x) = \frac{1}{2}(\log |\sigma_j^+(x)| - \log |\sigma_j^-(x)|).$$

On U , this is $\log |\sigma_j^+(x)|$. Its image is a full lattice in $\mathbb{R}^c$ of covolume R_U , where $R_U = 1$ if $c = 0$. Let w_K be the number of roots of unity in K , $h_{\text{rel}} = h_K/h_F$, $\kappa = |\ker(\text{Cl}(F) \rightarrow \text{Cl}(K))|$, and $L_F(s) = \zeta_K(s)/\zeta_F(s)$. The following identities give the mass of the norm-one units needed in the lattice construction.

Proposition 5.1. *We have that*

$$\kappa = I_U/2^{b-1}, \quad R_K/R_F = 2^{c-1}I_U R_U,$$

and consequently

$$(5.1) \quad \frac{w_K}{h_{\text{rel}}\kappa R_U} = \frac{2^{d-1}\pi^{b+c}}{\lambda^{d/2}L_F(1)}.$$

Here R_K, R_F are ordinary regulators, using twice the logarithm of absolute value at a complex place.

Proof. Dirichlet's theorem gives invariant and anti-invariant rational unit ranks $b + c - 1$ and c . The Herbrand quotient is multiplicative in exact sequences, equals one for finite groups, and is 2 or 1/2 on a trivial or sign copy of $\mathbb{Z}$. Applying this to the finite-index sum of the two eigenspace lattices gives the quotient 2^{b-1} , without requiring the integral unit lattice to split. Its degree-zero Tate group has size I_U , so $|H^1(\langle \iota \rangle, E_K)| = I_U/2^{b-1}$.

Capitulation injects into this H^1 , since a relation $\mathfrak{a}\mathcal{O}_K = (x)$ gives the cocycle $x/\iota x$ and a coboundary gives an invariant generator. Conversely, Hilbert 90 writes every norm-one unit as $x/\iota x$. The fractional ideal (x) is invariant. It descends from F , since exponents agree on split prime pairs and every finite ramification index is one. This proves surjectivity and the formula for κ .

The norm map on rational unit spaces is onto the invariant space, since the norm of an F -unit is its square. Its kernel has rank c . The logarithmic kernel in U consists of roots of unity by Kronecker's theorem. Every root of unity of K has norm one, since its norm is a torsion unit of F and is positive at a real place. Thus the logarithmic kernel has exactly w_K elements.

For the regulator identity, delete a coordinate over a real place of F in both weighted logarithmic unit spaces. The norm adds the two coordinates above a complex place of F . Replacing a pair (x, y) by $(x, x + y)$ has determinant one. On the kernel its first coordinate is $2l_j$, giving covolume $2^c R_U$. The logarithmic norm image has index $I_U/2$ in the logarithmic F -unit lattice. The torsion index is two, since -1 is not a norm at a complexifying real place. Multiplication of the kernel and image covolumes gives $R_K/R_F = 2^c R_U(I_U/2)$.

The analytic class-number formula [24, Chapter V, Theorem 2.4], with $w_F = 2$, gives

$$h_{\text{rel}} \frac{R_K}{R_F} = \frac{w_K \lambda^{d/2} L_F(1)}{2^{c+1} \pi^{b+c}}.$$

Substitution proves (5.1). $\square$

Choose finitely many primes $\mathfrak{p}$ of F split as $\mathfrak{P}\iota\mathfrak{P}$ in K , and integers $k_{\mathfrak{p}} \geq 0$. Set $P_0 = \prod_{\mathfrak{p}} (k_{\mathfrak{p}} + 1)$. The one-sided ideals $B_j = \prod \mathfrak{P}^{j_{\mathfrak{p}}}$ have a fiber of size at least $P_0/(\kappa h_{\text{rel}})$ in $\text{Cl}(K)/\text{im Cl}(F)$. If $B_j/B_{j_0} = (x)\mathfrak{a}_{\mathcal{O}_K}$, then

$$J_j/J_{j_0} = (x/\iota x), \quad J_j = \prod \mathfrak{P}^{j_{\mathfrak{p}}}(\iota\mathfrak{P})^{k_{\mathfrak{p}}-j_{\mathfrak{p}}}.$$

Thus $I = J_{j_0}^{-1}$ contains this many disjoint cosets $\beta_j U$ of norm-one elements. Their principal ideals distinguish the cosets, and

$$N_K(I) = \prod_{\mathfrak{p}} (N_F \mathfrak{p})^{-k_{\mathfrak{p}}}.$$

For uniform rational-prime data define

$$(5.2) \quad H = \sum_p \frac{k_p \log p}{e_p}, \quad J = \sum_p \frac{\log(k_p + 1)}{e_p f_p}.$$

Here e_p, f_p are the absolute indices at the selected primes, and all F -primes above them split in K/F . Then $N_K(I) = e^{-dH}$ and $P_0 = e^{dJ}$. Prime-ideal mixtures give the corresponding sums with a bounded rounding error, provided the finite list of exponents is fixed.

5.2. Unit Averaging and Translation. For $h \in \mathbb{R}^c$, rescale each complex pair by (e^{-h_j}, e^{h_j}) , leaving the b compact coordinates unchanged. Denote the resulting ideal lattice by Λ_h . Its determinant, independently of h , is

$$(5.3) \quad D = 2^{-d} \lambda^d e^{-dH}.$$

For any nonnegative integrable Φ on $\mathbb{R}^c$, averaging over a fundamental parallelepiped P_U of $l(U)$ gives exactly

$$(5.4) \quad \frac{1}{R_U} \int_{P_U} \sum_{\beta \in \bigcup_j \beta_j U} \Phi(l(\beta) - h) dh = \frac{q w_K}{R_U} \int_{\mathbb{R}^c} \Phi(u) du, \quad q \geq \frac{P_0}{\kappa h_{\text{rel}}}.$$

This follows from tiling and Tonelli's theorem for the full unit lattice. We use logarithmic Haar measure du and angular measures of total mass one.

Let Ω be bounded and invariant under rotations in each complex coordinate. Let V be its volume and let $\Phi(u)$ be its overlap with a displacement having compact moduli one and pair moduli (e^{u_j}, e^{-u_j}) . This function is nonnegative, bounded, and compactly supported in u . Parameterize additive translates by $B_h r$, $r \in [0, 1)^{2d}$, where B_h is a continuous basis matrix of Λ_h . The point count $N(h, r)$ and inherited ordered-edge count $E(h, r)$ are measurable countable sums of indicators. Averaging r by lattice tiling gives

$$\mathbb{E}_r N(h, r) = V/D, \quad \mathbb{E}_r E(h, r) = D^{-1} \sum_{\beta} \Phi(l(\beta) - h).$$

If a uniform point bound $N \leq N_{\max}$ holds for every h, r , all these quantities are integrable. Averaging also in h , and weighting parameter space by N , selects one pair (h, r) with $N > 0$ and

$$(5.5) \quad \frac{E(h, r)}{N(h, r)} \geq \frac{q w_K}{R_U} \frac{\int \Phi}{V}.$$

The vertex bound and the edge bound hold for the same rescaling and translate.

5.3. Uniform Lattice Counting. The real Euclidean dual of the unweighted complex ideal lattice is the coordinatewise complex conjugate of the image of $2\mathfrak{D}_K^{-1}I^{-1}$. Indeed its real pairing is the trace pairing divided by two. Reciprocal deformations invert on the dual. Consequently every nonzero dual vector satisfies

$$(5.6) \quad \prod_{i=1}^d |\xi_i| \geq \mu^d, \quad \mu = 2e^{H/2-\ell}, \quad \sum_i |\xi_i| \geq d\mu.$$

The product assertion follows by taking the square root of the absolute norm and using $|N_K(x)| \geq 1/(|\Delta_K|N_K(I))$ for $0 \neq x \in \mathfrak{D}_K^{-1}I^{-1}$. Here conjugation acts coordinatewise and need not agree with ι .

Suppose a smooth positive weight F on $\mathbb{C}^d$, formed as a product over the chosen one- and two-coordinate blocks, has normalized Fourier bound, for the convention $e^{-2\pi i x \cdot \xi}$,

$$|\widehat{F}(\xi)| / \int F \leq M^d \exp\left(-\sigma \sum_i |\xi_i|\right), \quad M \geq 1, \quad \sigma > 0.$$

Put $R = d\mu$ and use the norm $\|\xi\|_* = \sum_i |\xi_i|$. Every two distinct dual points are separated by at least R , so open balls of radius $R/2$ centered at them are disjoint. Comparing volumes inside the ball of radius $(j+3/2)R$ bounds the number of points in $jR \leq \|\xi\|_* < (j+1)R$ by $(2j+3)^{2d} \leq 5^{2dj}$. Thus the sufficient condition

$$(5.7) \quad \sigma\mu \geq \log M + 2\log 5 + 2$$

makes the nonzero normalized Fourier sum at most $\sum_{j \geq 1} e^{-2dj} < 1$. For Gaussian and polynomial Student profiles, we obtain

$$(5.8) \quad \sum_{x \in \Lambda_h + r} F(x) \leq 2 \int F/D,$$

uniformly in h, r . To justify Poisson summation, multiply F by $e^{-\varepsilon|x|^2}$. The result is Schwartz. Convolution with the normalized positive Fourier Gaussian multiplies the Fourier envelope by its exponential norm moment, which tends to one in each fixed dimension. The strict tail bound persists for sufficiently small ε . Schwartz Poisson summation and monotone convergence on the primal side prove (5.8). This limit is taken before d grows.

For $F_C(z) = e^{-ac|z|^2}$ with $0 < ac \leq 1$, one may use $2e^{-|\xi|}$ as a normalized Fourier envelope. For $F_C(z) = (1 + a|z|^2)^{-\nu}$, $\nu > 1$, Gaussian integration gives

$$\frac{\widehat{F}_C(\xi)}{\int F_C} = \frac{1}{\Gamma(\nu-1)} \int_0^\infty t^{\nu-2} e^{-t-\pi^2|\xi|^2/(at)} dt \leq 2^{\nu-1} e^{-\sqrt{2}\pi|\xi|/\sqrt{a}}.$$

The inequality follows from $t + L/t \geq t/2 + \sqrt{2L}$.

5.4. Compact and Complex-Pair Profiles. Fix $0 < \delta < 1$, $p = 2/(1+\delta)$, a positive radial f_C on $\mathbb{C}$, and a positive separately radial g_D on $\mathbb{C}^2$. Here p denotes the Lebesgue exponent. Define

$$A_C = \int f_C^p, \quad I_C = \int f_C(z) f_C(z+1) dz, \quad A_D = \int g_D^p,$$

$$I_D = \int_{\mathbb{R}} \int_{\mathbb{C}^2} g_D(x, y) g_D(x + e^u, y + e^{-u}) dx dy du,$$

$$J_C = \log I_C - (1+\delta) \log A_C, \quad J_D = \log I_D - (1+\delta) \log A_D.$$

Assume these integrals are finite and positive, the two endpoint log-energy variables have finite second moments, the total energy is bounded below and coercive, and (5.8) holds for the tensor weight $F = \prod f_C^p \prod g_D^p$ uniformly in reciprocal deformations. Section 7 verifies these hypotheses for our profiles. They give the following geometric criterion.

Proposition 5.2. *Suppose the preceding arithmetic data exist with $d \rightarrow \infty$, and $\log L_F(1) \leq dC$. With H, J, ℓ fixed up to bounded total rounding errors, set*

$$(5.9) \quad \mathcal{M}(\theta) = J - \delta H - (1/2 - \delta)\ell - C + (1 - \delta)\log 2 + (1 - \theta)\log \pi + (1 - 2\theta)J_C + \theta J_D.$$

If this margin has a fixed positive lower bound over all occurring signatures, there are arbitrarily large planar sets whose unordered unit-distance count divided by $|U|^{1+\delta}$ tends to infinity.

Proof. Write $V_C = -\log f_C$, $V_D = -\log g_D$. The compact endpoint law has density $f_C(z)f_C(z+1)/I_C$ with respect to dz . The pair endpoint law has density $g_D(x,y)g_D(x+e^u, y+e^{-u})/I_D$ with respect to $dx dy du$. These are probability laws. Reflection about the midpoint exchanges the two endpoints without changing either law. Let the first-endpoint energy means be v_C, v_D and set $v = (bv_C + cv_D)/d$. Take the common sublevel set

$$\Omega = \left\{ \sum V_C + \sum V_D \leq d(v + \eta) \right\}, \quad \eta > 0.$$

It is bounded. Its volume is at most $A_C^b A_D^c e^{pd(v+\eta)}$, and the pointwise weighted bound gives

$$N_{\max} \leq 2e^{dB_\eta}, \quad B_\eta = H + \log 2 - \ell + (1 - 2\theta)\log A_C + \theta \log A_D + pv + p\eta.$$

Independence across blocks, endpoint exchange symmetry, Chebyshev's inequality and a union bound put both endpoint energies within ηd of their means with probability at least $1/2$ for large d , uniformly in signature. Undoing the overlap density gives $\int \Phi \geq \frac{1}{2} I_C^b I_D^c e^{2d(v-\eta)}$. Equations (5.1) and (5.5) give ordered average degree at least $\frac{1}{4}e^{dA_\eta}$, where

$$\begin{aligned} A_\eta &= J + \log 2 + (1 - \theta)\log \pi - \ell/2 - C + (1 - 2\theta)\log I_C + \theta \log I_D \\ &\quad - (1 - 2\theta)\log A_C - \theta \log A_D + (2 - p)v - (2 + p)\eta. \end{aligned}$$

Since $p(1+\delta) = 2$, we have $A_\eta - \delta B_\eta = \mathcal{M}(\theta) - 4\eta$. The ordered ratio is at least $2^{-2-\delta} \exp(d(\mathcal{M} - 4\eta))$, apart from bounded rounding factors. Choose $4\eta < \inf \mathcal{M}$. A compact coordinate comes from a real place of F . Projection to that embedding is injective on differences of affine ideal points and sends all distinguished norm-one displacements to unit vectors. Division by two gives the unordered conclusion. Since $u(U) \leq |U|^2/2$ and $0 < \delta < 1$, the ratio $u(U)/|U|^{1+\delta}$ can tend to infinity only if $|U| \rightarrow \infty$. This gives a sequence of cardinalities, with no claim at every sufficiently large cardinality. $\square$

6. WEIGHTED WINDOWS AT SPLIT FINITE PLACES

We extend the geometric criterion by placing locally constant weights at the selected finite primes. Keep the notation K/F , $d = b + 2c$, $\theta = c/d$, λ and $\ell = \log \lambda$ of Section 5, where $b > 0$ and K/F is unramified at all finite places. Put $D_0 = (\lambda/2)^d$, and keep the archimedean profiles f_C, g_D and functionals J_C, J_D defined there.

6.1. The Local Functional. Let L be a nonarchimedean local field with valuation ring $\mathcal{O}$, uniformizer π , and residue cardinality Q . Additive Haar measure gives $\mathcal{O}^2$ mass one. Multiplicative Haar measure du on $\mathcal{O}^\times$ has mass one, and $n \in \mathbb{Z}$ has counting measure. Define

$$s(n, u) = (\pi^n u, \pi^{-n} u^{-1}), \quad Tg(z) = \sum_{n \in \mathbb{Z}} \int_{\mathcal{O}^\times} g(z + s(n, u)) du.$$

For a nonnegative, locally constant, compactly supported g , put

$$(6.1) \quad A(g) = \int g^p, \quad Z(g) = \int g Tg, \quad \mathcal{F}_\delta(g) = \frac{Z(g)}{A(g)^{1+\delta}}, \quad p = \frac{2}{1+\delta}, \quad 0 < \delta < 1.$$

Assume $g \neq 0$ and $Z(g) > 0$. Only finitely many valuation labels contribute, since both coordinates of $s(n, u)$ lie in a fixed compact difference set. The unit average has mass one.

For any integer a , put $B_a = \pi^{-a}\mathcal{O}$. Ball intersection gives

$$(6.2) \quad \int 1_{B_a \times B_b} T 1_{B_c \times B_e} = Q^{\min(a,c) + \min(b,e)} [\max(a,c) + \max(b,e) + 1]_+.$$

Two translated balls intersect with the smaller volume exactly when the displacement lies in the larger ball. The first displacement must have valuation at least $-\max(a,c)$ and the second at least $-\max(b,e)$. These conditions say $-\max(a,c) \leq n \leq \max(b,e)$. Every permitted unit contributes the same intersection volume. This also proves the formula for negative ball indices.

Put $S_0 = \mathcal{O}$, $S_i = B_i \setminus B_{i-1}$ for $i \geq 1$, and $m_0 = 1$, $m_i = Q^i - Q^{i-1}$. For $k \geq 0$, let g have value w_{ij} on $S_i \times \pi^{-k}S_j$ for $0 \leq i \leq L_x$, $0 \leq j \leq L_y$, and be zero elsewhere. The weights are nonnegative. With

$$D_{ir} = \begin{cases} m_{\min(i,r)}, & i \neq r, \\ 0, & i = r = 0, \\ im_i - Q^{i-1}, & i = r \geq 1, \end{cases}$$

define

$$K_{(i,j),(r,t)} = (k+1)m_im_j\mathbf{1}_{i=r,j=t} + m_jD_{ir}\mathbf{1}_{j=t} + m_iD_{jt}\mathbf{1}_{i=r}.$$

Then

$$(6.3) \quad A(g) = Q^k \sum_{i,j} m_im_j w_{ij}^p, \quad Z(g) = Q^k w^T K w.$$

To verify the second formula, take the four shell differences of (6.2). The constant $k+1$ gives the diagonal term. The double difference of $\max(i,r)Q^{\min(i,r)}$ is D_{ir} , whereas that of $Q^{\min(j,t)}$ is $m_j\mathbf{1}_{j=t}$. The remaining term is symmetric. Since $k \geq 0$, the positive-part sign can be removed in all these differences. The formula holds in residue characteristic two and for ramified $L/\mathbb{Q}_p$.

These profiles are separately invariant under multiplication by units and have the additive period

$$(6.4) \quad C = \mathcal{O} \times \pi^{-k}\mathcal{O}, \quad |C| = Q^k,$$

irrespective of the number of shells. In particular $w_{00} = 1$ and all other weights zero recover $\mathcal{F}_\delta = (k+1)Q^{-\delta k}$.

For product weights $w_{ij} = a_ib_j$, define

$$P_a = \sum_i m_i a_i^p, \quad S_a = \sum_i m_i a_i^2, \quad R_a = \sum_{i,r} a_i D_{ir} a_r,$$

and define P_b, S_b, R_b similarly. Formula (6.3) becomes

$$(6.5) \quad A(g) = Q^k P_a P_b, \quad Z(g) = Q^k [(k+1)S_a S_b + R_a S_b + S_a R_b].$$

Thus symmetric product weights need only one vector. The one-shell choice $(a_0, a_1) = (b_0, b_1) = (1, x)$ has $S = 1 + (Q-1)x^2$ and $R = 2x + (Q-2)x^2$, so its log gain over the hard rectangle is

$$(6.6) \quad \log \frac{(k+1)S^2 + 2SR}{k+1} - 2(1+\delta) \log(1 + (Q-1)x^p).$$

Its right derivative at zero is $4/(k+1) > 0$. For the numerical certificate we use the full finite functional.

6.2. Finite Periods and Lattice Counting. Let S now denote a finite set of F -primes split in K . At each $v \in S$ identify the two completions with $L_v = F_v$, compatibly with ι , and choose a common uniformizer. Put

$$E_S = \prod_{v \in S} (L_v \times L_v), \quad \Gamma = \mathcal{O}_{K,S} \subset \mathbb{C}^d \times E_S.$$

Here $\mathcal{O}_{K,S}$ consists of the elements integral outside the selected prime pairs. For a rectangular compact open subgroup $C_f = \prod_v (\pi_v^{a_v} \mathcal{O}_v \times \pi_v^{b_v} \mathcal{O}_v)$, the elements of Γ with finite component in C_f form the fractional ideal

$$L = \prod_v \mathfrak{P}_v^{a_v} (\iota \mathfrak{P}_v)^{b_v}, \quad N_K(L) = |C_f|^{-1}, \quad \text{covol}(L) = D_0/|C_f|.$$

Every C_f -coset in E_S contains a finite component of a Γ -point. To see this, enclose the coset and C_f in larger local balls and form the corresponding fractional ideal $J \supset L$. After clearing a common denominator, the Chinese Remainder Theorem identifies J/L with its local quotients at the selected prime ideals. This requires no principality hypothesis, and each finite coset has an archimedean fiber equal to a translate of L .

If P_L is an archimedean fundamental parallelepiped, then $P_L \times C_f$ is a fundamental domain for Γ , of Haar volume D_0 . First subtract a Γ -element to put the finite component in C_f , then subtract an L -element to put the infinite component in P_L . The same argument proves uniqueness and discreteness. Reciprocal archimedean deformation B_h gives $\Gamma_h = (B_h, \text{id})\Gamma$ with the same covolume.

Write $H_C = d^{-1} \log |C_f|$. The codifferent argument in (5.6) applied to L gives the product minimum $[2e^{H_C/2-\ell}]^d$ on every deformed dual lattice. Consequently the archimedean Fourier condition sufficient for this section is

$$(6.7) \quad \sigma 2e^{H_C/2-\ell} \geq \log M + 2 \log 5 + 2,$$

using the envelope and pointwise regularity of Section 5. It gives a bound $2 \int F_\infty / (D_0/|C_f|)$ for the sum of $F_\infty = \prod f_C^p \prod g_D^p$ on every archimedean fiber. Summing against any nonnegative, compactly supported, C_f -periodic function F_f yields

$$(6.8) \quad \sum_{\gamma \in \Gamma_h + w} F_\infty(\gamma_\infty) F_f(\gamma_f) \leq \frac{2}{D_0} \int F_\infty \int F_f.$$

The values of F_f on its finitely many nonzero cells sum to $(\int F_f)/|C_f|$, so the estimate depends on the period volume. Local reciprocal diagonal rescaling replaces C_f by a rectangular period of the same volume. Hence the bound remains uniform after such rescalings, including those that do not preserve Γ .

6.3. Class Selection and Unit Averaging. Let $\Omega \subset \mathbb{C}^d \times E_S$ be bounded, have compact finite support, and be invariant under all coordinate rotations and local reciprocal unit multiplications. Denote its overlap with a norm-one step of archimedean pair-log coordinates $u \in \mathbb{R}^c$ and finite valuation labels $n \in \mathbb{Z}^S$ by $\Phi(u, n)$. Invariance removes all phase and unit choices. Only finitely many labels occur, and the u -support is bounded. Form the weights

$$W(n) = \int_{\mathbb{R}^c} \Phi(u, n) du.$$

The quotient $\text{Cl}(K)/\text{im Cl}(F)$ has order κh_{rel} . Map n to the class of $A(n) = \prod_v \mathfrak{P}_v^{n_v}$. One class has total weight at least $\sum_n W(n)/(\kappa h_{\text{rel}})$. Suppose this sum is positive and choose a positive-weight reference label n_0 in that class. For every selected label, write $A(n)/A(n_0) = (x_n) \mathfrak{a}_n \mathcal{O}_K$. Then

$$\beta_n = x_n / \iota(x_n), \quad N_{K/F} \beta_n = 1, \quad (\beta_n) = \prod_v \mathfrak{P}_v^{n_v - n_{0,v}} (\iota \mathfrak{P}_v)^{-n_v + n_{0,v}}.$$

Then $\beta_n \in \mathcal{O}_{K,S}$ and the full set of norm-one generators of this ideal is $\beta_n U$. Distinct labels give disjoint cosets.

Let α_0 be the identity at infinity with finite coordinates $(\pi_v^{n_{0,v}}, \pi_v^{-n_{0,v}})$, and put $\Omega' = \alpha_0^{-1} \Omega$. This preserves Haar and period volumes. The overlap with the displacement $(B_h \beta, \beta_f)$, for $\beta \in \beta_n U$, is

$\Phi(l(\beta) - h, n)$. Since the overlap is independent of units and phases, correlations between local units of global generators and their archimedean data do not affect it. Tiling by the full unit lattice gives

$$\frac{1}{R_U} \int_{P_U} \sum_{n \text{ selected}} \sum_{\beta \in \beta_n U} \Phi(l(\beta) - h, n) dh = \frac{w_K}{R_U} \sum_{n \text{ selected}} W(n).$$

For each fixed field and bounded h -domain, only finitely many unit-log lattice points meet the bounded overlap support, and the logarithmic kernel has w_K elements. Tonelli also justifies the sums directly, since their terms are nonnegative. We need no distribution hypothesis on the local unit images.

Jointly average h and additive translations over a fundamental domain of Γ_h , then weight parameter space by its point count. The translation averages are $|\Omega|/D_0$ for vertices and the sum of overlaps divided by D_0 for ordered edges. Hence some pair of transformations gives a nonempty point set satisfying

$$(6.9) \quad \frac{E}{N} \geq \frac{w_K}{\kappa h_{\text{rel}} R_U} \frac{\sum_n \int \Phi(u, n) du}{|\Omega|}.$$

The uniform bound (6.8) holds for this same pair. Choose the parameter domains using a continuously deformed basis and Haar measure on the finite period. Countable indicator sums and Tonelli justify the averages, including sublevel boundaries, since the exponential majorant is pointwise and additive tiling holds for measurable indicators. The class is chosen from the overlaps of the final window, so concentration will be applied before class selection.

6.4. The Finite-Place Transfer. For every selected place choose a nonzero, nonnegative, locally constant, compactly supported g_v with a rectangular additive period C_v and the unit invariance above, and with $Z_v = Z(g_v) > 0$. Suppose their number is $O(d)$ and their profiles belong to a fixed finite list of local types. The finite shell profiles above meet these hypotheses. Write $A_v = A(g_v)$ and $\mathcal{F}_{\delta, v} = Z_v/A_v^{1+\delta}$. These local functionals give the following extension of Proposition 5.2.

Proposition 6.1. *Suppose the field degrees tend to infinity, $\log L_F(1) \leq dC$, and (6.7) holds with $C_f = \prod_v C_v$. Assume the archimedean energy and moment hypotheses of Proposition 5.2. If*

$$(6.10) \quad \begin{aligned} \mathcal{M}_{\text{fin}}(\theta) = & \frac{1}{d} \sum_{v \in S} \log \mathcal{F}_{\delta, v} - (1/2 - \delta)\ell - C + (1 - \delta) \log 2 + (1 - \theta) \log \pi \\ & + (1 - 2\theta)J_C + \theta J_D \end{aligned}$$

has a fixed positive lower bound, there are planar sets of cardinality tending to infinity whose unordered unit-distance count divided by $n^{1+\delta}$ tends to infinity.

Proof. The total profile $f = \prod f_C \prod g_D \prod_v g_v$ has

$$A = \int f^p = A_C^b A_D^c \prod_v A_v, \quad Z = I_C^b I_D^c \prod_v Z_v.$$

Use the independent normalized overlap probability laws of all these blocks. The two endpoint energy distributions agree by reflection and the stated symmetries. Let $d\bar{v}$ be the sum of first-endpoint means for energy $-\log f$. On each finite factor, the positive support is a finite union of additive cells with a positive minimum weight, so its endpoint energy is bounded. The total variance is $O(d)$.

On the positive support put $\Omega = \{-\log f \leq d(\bar{v} + \varepsilon)\}$, for $\varepsilon > 0$. Coercivity makes it bounded. Its finite support is compact and it has all the required periods and invariances. Exponential majorization and (6.8), also after the reference-label rescaling, give

$$|\Omega| \leq A e^{pd(\bar{v} + \varepsilon)}, \quad N \leq 2A e^{pd(\bar{v} + \varepsilon)} / D_0.$$

Chebyshev's inequality and a union bound put both endpoint energies in $[d(\bar{v} - \varepsilon), d(\bar{v} + \varepsilon)]$ with probability at least $1/2$ for large d . Undoing the overlap density on this event gives

$$\sum_n \int \Phi(u, n) du \geq \frac{1}{2} Z e^{2d(\bar{v} - \varepsilon)}.$$

Since $p(1 + \delta) = 2$, (6.9) implies

$$\frac{E}{N^{1+\delta}} \geq \frac{w_K D_0^\delta}{2^{1+\delta} \kappa h_{\text{rel}} R_U} \frac{Z}{A^{1+\delta}} e^{-4\varepsilon d}.$$

Substituting (5.1), the ordered ratio is at least $2^{-2-\delta} e^{d(\mathcal{M}_{\text{fin}} - 4\varepsilon)}$. Choose $4\varepsilon < \inf \mathcal{M}_{\text{fin}}$ and divide by two for unordered edges. Projection to a compact coordinate is injective on the field points and sends the norm-one displacements to unit vectors. The upper bound $N^2/2$ for unordered edges forces $N \rightarrow \infty$. $\square$

If a rational prime has actual indices e_p, f_p and all its F -places split in K/F , there are $d/(e_p f_p)$ such places. A common local profile contributes $\log \mathcal{F}_{\delta,p}/(e_p f_p)$ to (6.10). Hard rectangles give $\log \mathcal{F}_{\delta,p} = \log(k_p + 1) - \delta k_p f_p \log p$, so their sum is exactly $J - \delta H$. Adding finite shells with period (6.4) preserves $H_C = \sum_p k_p \log p / e_p = H$. Hence the finite-place functional replaces the ideal-choice term and leaves the Fourier condition unchanged.

7. ARCHIMEDEAN PROFILES

For a pair of complex places, set

$$t = (1 + a|z|^2)^{-1}, \quad u = (1 + a|w|^2)^{-1}, \quad g_D(z, w) = t^s u^s P(t, u),$$

where

$$s = 1.14600585, \quad a = 0.0000128633241758,$$

and

$$P(t, u) = \sum_{i,j=0}^3 B_{ij} b_i^3(t) b_j^3(u), \quad b_i^m(t) = \binom{m}{i} t^i (1-t)^{m-i},$$

with

$$(7.1) \quad B = \begin{pmatrix} 1 & 2.4129438977 & 2.7721799210 & 6.6857798061 \\ 2.4129438977 & 3.8532983888 & 4.6371750119 & 8.1307168183 \\ 2.7721799210 & 4.6371750119 & 5.4628959084 & 9.5512647588 \\ 6.6857798061 & 8.1307168183 & 9.5512647588 & 13.8659430610 \end{pmatrix}.$$

All finite decimals are exact rationals. The Bernstein basis is nonnegative and sums to one, so P is bounded above and below by positive constants on the real square. We prove the hypotheses of Proposition 6.1 for these profiles and derive finite bounds for their functionals. Section 8 evaluates the bounds at the fixed rational parameters.

7.1. The Gaussian and the Pair Convolution. The compact Gaussian $f_C^p = e^{-a_C|z|^2}$ gives

$$(7.2) \quad J_C = \log(p/2) + \delta \log(a_C/\pi) - a_C/(2p), \quad a_C = 2p\delta$$

where the displayed scale maximizes the Gaussian functional. For our choice of δ , we have $a_C < 1$. The slope of the general margin as a function of the signature is $-\log \pi - 2J_C + J_D$, and depends on the chosen profile.

For the pair calculation, let $\alpha, \alpha' > 0$ with $A = \alpha + \alpha' - 1 > 0$. The Laplace identity

$$(1 + |z|^2)^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty r^{\alpha-1} e^{-r-r|z|^2} dr$$

and Gaussian integration give

$$(7.3) \quad \int_{\mathbb{C}} (1 + |z|^2)^{-\alpha} (1 + |z + h|^2)^{-\alpha'} dz = \frac{\pi}{A} \mathbb{E}(1 + T(1 - T)|h|^2)^{-A}, \quad T \sim \text{Beta}(\alpha, \alpha').$$

After integrating z , change the two positive Laplace variables to their sum r and ratio T . Completing the square gives the term $rT(1 - T)|h|^2$, and integration in r gives the Beta expectation. Tonelli justifies the interchanges, since the integrands are nonnegative.

For $A, B > 0$ define

$$Q_{AB}(b) = \int_{\mathbb{R}} (1 + be^{2v})^{-A} (1 + be^{-2v})^{-B} dv, \quad b > 0.$$

The two coordinates in the pair convolution reduce to this integral with

$$b = a\sqrt{T(1 - T)T'(1 - T')} \leq a/4.$$

This follows by translating v until the coefficients of e^{2v} and e^{-2v} agree, which has Jacobian one. The exponential representation is

$$\int_{\mathbb{R}} e^{-a(Re^{2v} + R'e^{-2v})} dv = K_0(2a\sqrt{RR'}), \quad K_0(x) = \int_0^\infty e^{-x \cosh t} dt,$$

where the substitution $t = 2v + \frac{1}{2} \log(R/R')$ and evenness cancel their factors of two.

Splitting the Q_{AB} integral at zero yields

$$(7.4) \quad \left| Q_{AB}(b) - \log(1/b) + \frac{\psi(A) + \psi(B) + 2\gamma_E}{2} \right| \leq (A + B)b/2.$$

Deleting the smaller factor on one half leaves $\frac{1}{2} \int_b^\infty (1 + x)^{-A} dx/x$. Its difference from its logarithmic constant lies between zero and $Ab/2$, by $0 \leq 1 - (1 + x)^{-A} \leq Ax$. On that half the factor removed costs at most $Bb/2$, since $1 - (1 + be^{-2v})^{-B} \leq Bbe^{-2v}$. The other half interchanges A and B . Thus the total positive error from the two tails and the total loss from the factors removed are both at most $(A + B)b/2$, proving the bound. The logarithmic constant follows from [27, Equation (5.9.16)],

$$\int_0^1 \frac{(1 - x)^{A-1} - 1}{x} dx = -\psi(A) - \gamma_E.$$

We use $\psi = \Gamma'/\Gamma$ and Euler's constant γ_E . For later formulas write

$$(7.5) \quad C_s = -2\psi(s) + \psi(2s) + (2s - 1)^{-1} - \gamma_E.$$

7.2. Endpoint Moments and the Fourier Bound. The compact Gaussian has finite positive mass and overlap, and every polynomial moment under its endpoint law is finite. Since P is bounded above and below by positive constants, the pair mass is finite for $sp > 1$. Formula (7.3) and (7.4) show that the pair overlap is finite: the Beta variables have positive parameters and finite logarithmic moments, and $b \leq a/4$. Both integrals are positive. The energy $-\log g_D = s \log(1 + a|z|^2) + s \log(1 + a|w|^2) - \log P(t, u)$ is bounded below and coercive, so its sublevel sets are compact.

Finite endpoint-energy second moments hold for every positive polynomial Student profile $t^s u^s P(t, u)$ when P is bounded above and below by positive constants on the square and $sp > 1$. Choose $0 < \epsilon < \min(s, 2s - 1)$ and absorb each logarithmic square into a factor $(1 + a|z|^2)^\epsilon$. Formula (7.3) then has positive Beta parameters and positive hyperbola exponents. The bound in (7.4) is a constant plus $|\log b| + b(A + B)/2$, which is integrable against the Beta densities. This proves the second-moment condition, including $s = 1$.

For the Fourier bound, write $P(t, u) = \sum B_{ij} b_i^m(t) b_j^m(u)$ with $B_{ij} > 0$. For a tube width $0 < \rho < 1$, put $\omega = (\rho + \rho^2)/(1 - \rho^2)$ and

$$E_{rs} = \binom{m}{r} \binom{m}{s} \max |\Delta_1^r \Delta_2^s B|, \quad q_0 = \frac{\sum_{r+s>0} E_{rs} \omega^{r+s}}{\min B}.$$

If $q_0 < 1$, the Bernstein derivative formula puts P in the right half-plane throughout the product complex tube. Contour shifting then gives, with $\beta = sp$,

$$|\widehat{g_D^p}(\xi_z, \xi_w)|/A_D \leq (1 - \rho^2)^{-2\beta} (1 + q_0)^p e^{-2\pi\rho(|\xi_z| + |\xi_w|)/\sqrt{a}}.$$

Here the tube is taken after the scaling $x = \sqrt{a}z$ in each complex coordinate, and its imaginary displacement has Euclidean norm at most ρ . If $D(x, y) = 1 + \sum_j (x_j + iy_j)^2$, then

$$|D(x, y)| \geq (1 - \rho^2)(1 + |x|^2), \quad |D(x, y)^{-1} - (1 + |x|^2)^{-1}| \leq \omega.$$

The latter follows by bounding the numerator by $2|x|\rho + \rho^2$ and the denominator by $(1 - \rho^2)(1 + |x|^2)^2$. The Bernstein derivative identity bounds the coefficient of the (r, s) Taylor term by E_{rs} , uniformly at real $(t, u) \in [0, 1]^2$. Hence the complex value of P differs from its positive real value by at most $q_0 P(t, u)$. We can define P^p holomorphically in the tube and have $|P_{\text{shifted}}|^p \leq (1 + q_0)^p P(t, u)^p$. The displayed bound on D supplies the other factor $(1 - \rho^2)^{-2sp}$. Shift each real Fourier contour by a vector of length $\rho/\sqrt{a}$ opposite its frequency. The shifted majorant is integrable since $sp > 1$, so truncation and dominated convergence justify the shift. This proves the Fourier envelope. The tube condition $q_0 < 1$, in addition to positivity on the real square, gives explicit M, σ in (5.7).

With the compact Gaussian (7.2), a sufficient global choice in (5.7) is

$$(7.6) \quad M = \max\{2, \sqrt{K_D}\}, \quad \sigma = \min\{1, 2\pi\rho/\sqrt{a}\}, \quad K_D = (1 - \rho^2)^{-2sp} (1 + q_0)^p.$$

There are b one-coordinate blocks and c two-coordinate blocks, so their prefactors are bounded by $M^{b+2c} = M^d$. The maximum and minimum give a bound uniform in the signature.

7.3. Finite Bounds for Polynomial Profiles. Let $s \geq 1$, $\beta = sp > 1$, $\eta_0 = 2s - 1$, and

$$g_D(z, w) = t^s u^s P(t, u), \quad P(t, u) = \sum_{i,j} d_{ij} t^i u^j > 0 \quad (0 \leq t, u \leq 1).$$

Put

$$\begin{aligned} A_{ik} &= \eta_0 + i + k, \quad B_{jl} = \eta_0 + j + l, \quad w_{ijkl} = \frac{d_{ij} d_{kl}}{A_{ik} B_{jl}}, \quad b_0 = \sum w_{ijkl} > 0, \\ c(x, y) &= \frac{1}{x + y - 1} + \frac{\psi(x + y - 1) - \psi(x) - \psi(y) - \gamma_E}{2}, \\ \kappa_P &= \frac{\sum w_{ijkl} [c(s + i, s + k) + c(s + j, s + l)]}{b_0}. \end{aligned}$$

Here $\pi^2 b_0 = \int_{\mathbb{C}^2} g_D^2$ at scale $a = 1$. Let $z_0 = a^2/16$ and assume, for every cross with nonzero coefficient,

$$(7.7) \quad A_{ik} B_{jl} z_0 < 1, \quad 2 \log(4/a) \geq H_{\lceil A_{ik} \rceil - 1} + H_{\lceil B_{jl} \rceil - 1}, \quad \log(4/a) > 1/2,$$

where $H_0 = 0$. Define

$$E_- = \log(4/a) \sum_{w_{ijkl} < 0} (-w_{ijkl}) \frac{A_{ik} B_{jl} z_0}{1 - A_{ik} B_{jl} z_0}.$$

Then

$$(7.8) \quad I_D \geq \pi^2 a^{-2} b_0 [\log(1/a) + \kappa_P - E_-/b_0].$$

The bracket must be positive before taking its logarithm.

To prove this bound, recall that for $A, B \geq 1$,

$$Q_{AB}(b) = \int_{\mathbb{R}} (1 + be^{2u})^{-A} (1 + be^{-2u})^{-B} du.$$

For $0 < b < 1$, its convergent logarithmic series is

$$Q_{AB}(b) = \sum_{n \geq 0} \frac{(A)_n (B)_n}{(n!)^2} b^{2n} \left[\log(1/b) + \psi(n+1) - \frac{\psi(A+n) + \psi(B+n)}{2} \right].$$

To verify the series, substitute $x = be^{2u}$ and then $t = x/(x+b^2)$. This expresses $2Q_{AB}(\sqrt{z})$ as

$$\int_0^1 t^{B-1} (1-t)^{A-1} (1-(1-z)t)^{-A} dt.$$

Differentiation and integration by parts give $z(1-z)y'' + [1 - (A+B+1)z]y' - AB y = 0$. Substituting the displayed series verifies its coefficients, using $a_{n+1}/a_n = (n+A)(n+B)/(n+1)^2$ and the digamma recurrence. The coefficients have polynomial growth, so the series and its fixed derivatives converge locally uniformly for $0 < z < 1$. Its leading logarithm and constant agree with the elementary estimate (7.4). To see uniqueness with these two leading terms, let $\phi(z) = \sum a_n z^n$, so $\phi(0) = 1$ and $\phi > 0$ on $[0, 1]$. For the difference v of two such solutions, reduction of order gives $(v/\phi)' = C/[z(1-z)^{A+B}\phi^2]$. Thus $v/\phi = C \log z + D + o(1)$, and the matching leading terms force $C = D = 0$.

The digamma recurrence and monotonicity imply

$$-H_{[A]-1} - H_{[B]-1} \leq 2\psi(n+1) - \psi(A+n) - \psi(B+n) \leq 0.$$

Under the threshold in (7.7), every term with $n \geq 1$ is nonnegative. Since $(A)_n (B)_n / (n!)^2 \leq (AB)^n$, summation gives

$$0 \leq Q_{AB}(b) - \log(1/b) + \frac{1}{2}[\psi(A) + \psi(B) + 2\gamma_E] \leq \frac{ABb^2 \log(1/b)}{1 - ABb^2}.$$

The right side increases for $b \leq a/4$ under the stated conditions. Apply this inside the two Beta expectations from (7.3), where $b = a\sqrt{T(1-T)T'(1-T')} \leq a/4$. Their logarithmic expectations give $c(x, y)$. For positive cross coefficients we discard the nonnegative remainder. For negative cross coefficients we subtract its upper bound. This proves (7.8), allowing either sign of monomial coefficient.

To retain cancellation between monomial coefficients, we collect exact terms before bounding the signed remainder. For $n \geq 0$, define

$$K_n(i, k) = \frac{(s+i)_n (s+k)_n}{n!(\eta_0 + i + k + n)_{n+1}}, \quad S_x(j) = \sum_{h=0}^{j-1} \frac{1}{x+h},$$

with $S_x(0) = 0$, and put

$$\begin{aligned} r_n(i, k) &= -\eta_0^{-1} + H_n/2 - S_{\eta_0}(i+k+n)/2 + S_{\eta_0}(i+k+2n+1) \\ &\quad - [S_s(i+n) + S_s(k+n)]/2, \quad R_n(i, k) = K_n(i, k)r_n(i, k), \end{aligned}$$

$$B_n = \sum d_{ij} d_{kl} K_n(i, k) K_n(j, l), \quad V_n = \sum d_{ij} d_{kl} [R_n(i, k) K_n(j, l) + K_n(i, k) R_n(j, l)].$$

For rational parameters these are rational numbers. Under (7.7), for every integer $m \geq 0$,

$$\begin{aligned} (7.9) \quad \frac{I_D}{\pi^2 a^{-2}} &\geq \sum_{n=0}^m a^{2n} \{B_n [\log(1/a) + C_s] + V_n\} - E_{-,m}, \\ E_{-,m} &= \log(4/a) \sum_{w_{ijkl} < 0} (-w_{ijkl}) \frac{(A_{ik} B_{jl} z_0)^{m+1}}{1 - A_{ik} B_{jl} z_0}. \end{aligned}$$

Thus $m = 1$ retains the exact a^2 term and leaves a signed error of order $a^4 \log(1/a)$ for a fixed polynomial. The displayed right-hand side is a lower bound for the overlap and gives no upper bound.

For a cross with exponents $\alpha = s + i$, $\alpha' = s + k$ and $A = \alpha + \alpha' - 1$, the coordinate factor at order n is

$$\frac{(A)_n}{An!} \mathbb{E}[(T(1-T))^n] = \frac{(\alpha)_n(\alpha')_n}{n!(A+n)_{n+1}} = K_n(i, k), \quad T \sim \text{Beta}(\alpha, \alpha').$$

The logarithmic moment uses the tilted distribution $\text{Beta}(\alpha + n, \alpha' + n)$, giving the coordinate constant

$$\frac{1}{2}\psi(n+1) - \frac{1}{2}\psi(A+n) - \frac{1}{2}[\psi(\alpha+n) + \psi(\alpha'+n)] + \psi(A+2n+1) = C_s/2 + r_n(i, k).$$

The digamma recurrence proves the rational expression for r_n . The remainder of each Q_{AB} after index m is nonnegative and bounded by $\log(1/b)(ABb^2)^{m+1}/(1-ABb^2)$. Monotonicity for $b \leq a/4$ under (7.7) proves (7.9). We also have $B_n > 0$, since K_n has the Gram representation

$$K_n(i, k) = \frac{(s+i)_n(s+k)_n}{(n!)^2} \int_0^1 t^{\eta_0+n-1} (1-t)^n t^i t^k dt.$$

It is positive definite on any finite collection of distinct monomials, and its tensor square makes B_n the squared norm of a nonzero polynomial. Individual signed crosses can still be negative.

For the mass, let T, U be independent $\text{Beta}(\beta - 1, 1)$ variables. Radial substitution gives

$$(7.10) \quad A_D = \frac{\pi^2 a^{-2}}{(\beta - 1)^2} \mathcal{A}, \quad \mathcal{A} = \mathbb{E}P(T, U)^p.$$

For a rational positive Bernstein representation, first clear a common denominator. Since scalar multiplication of P cancels from J_D , normalize it to $P = 1 + V$ with $|V| \leq r < 1$ by dividing by half the sum of the smallest and largest Bernstein coefficient. Then, for $N \geq 2$,

$$(7.11) \quad \left| \mathcal{A} - \sum_{k=0}^N \binom{p}{k} \mathbb{E}V^k \right| \leq \frac{| \binom{p}{N+1} | r^{N+1}}{1-r}.$$

Indeed the binomial series is uniformly convergent on $[-r, r]$ and the absolute binomial coefficients decrease after index two, since $1 < p < 2$. If $\beta - 1 = A/C$ is positive rational, the monomial moments $\mathbb{E}T^j = A/(A + Cj)$ make every coefficient in the truncated sum exactly rational. We calculate them by integer polynomial convolution and contraction. Since the mass has a negative logarithmic coefficient in the functional, its upper endpoint gives a lower bound for the score.

For polynomials with a large ratio of maximum to minimum Bernstein coefficient, the following local version has a smaller tail. Partition $[0, 1]^2$ into rational rectangles, put $q = \beta - 1 > 0$, and on $\mathcal{R} = [l, r] \times [b, t]$ use coordinates $x = (T - l)/(r - l)$ and $y = (U - b)/(t - b)$. Restrict the polynomial exactly and express it in the Bernstein basis in x, y . If its coefficient bounds are $0 < m_{\mathcal{R}} \leq M_{\mathcal{R}}$, set

$$h_{\mathcal{R}} = (m_{\mathcal{R}} + M_{\mathcal{R}})/2, \quad \rho_{\mathcal{R}} = \frac{M_{\mathcal{R}} - m_{\mathcal{R}}}{M_{\mathcal{R}} + m_{\mathcal{R}}} < 1, \quad P = h_{\mathcal{R}}(1 + W_{\mathcal{R}}).$$

The Bernstein convex-hull property gives $|W_{\mathcal{R}}| \leq \rho_{\mathcal{R}}$ throughout the rectangle. Define the unnormalized one-dimensional moments

$$(7.12) \quad \begin{aligned} M_i(l, r) &= \int_l^r qv^{q-1} \left(\frac{v-l}{r-l} \right)^i dv \\ &= \frac{q}{(r-l)^i} \sum_{j=0}^i \binom{i}{j} (-l)^{i-j} \frac{r^{q+j} - l^{q+j}}{q+j}. \end{aligned}$$

The identity follows by expanding $(v - l)^i$. The convention $0^q = 0$ is valid because $q > 0$. If $W_{\mathcal{R}}^k = \sum c_{ij}^{(k)} x^i y^j$, put

$$T_{\mathcal{R},N} = h_{\mathcal{R}}^p \sum_{k=0}^N \binom{p}{k} \sum_{i,j} c_{ij}^{(k)} M_i(l, r) M_j(b, t).$$

Then for $N \geq 2$,

$$(7.13) \quad \left| \mathcal{A} - \sum_{\mathcal{R}} T_{\mathcal{R},N} \right| \leq \sum_{\mathcal{R}} h_{\mathcal{R}}^p \frac{\binom{p}{N+1} |\rho_{\mathcal{R}}|^{N+1}}{1 - \rho_{\mathcal{R}}} M_0(l, r) M_0(b, t).$$

This follows by integrating the uniform binomial-tail bound against the nonnegative density on each rectangle. Polynomial restriction, conversion, and powers have rational coefficients and may be calculated by integer convolution after clearing denominators. The endpoint powers in (7.12) and $h_{\mathcal{R}}^p$ are enclosed using outward-rounded logarithms and exponentials, since they need not be rational. Exact rational algebra followed by interval contraction preserves the bound when the moment formula has cancellation. For symmetric P and the same partition on both axes, off-diagonal rectangles may be counted twice and diagonal rectangles once. Otherwise all rectangles must be included separately.

Combining (7.8) and (7.10) gives

$$(7.14) \quad J_D \geq \log b_0 + 2(1 + \delta) \log(\beta - 1) - 2\delta \log \pi - (1 + \delta) \log \mathcal{A} + 2\delta \log a \\ + \log[\log(1/a) + \kappa_P - E_-/b_0].$$

More generally, if $Z_* > 0$ is the lower expression in (7.9) and $\mathcal{A}^*$ is an upper bound from (7.13), we obtain the bound used in the certificate,

$$(7.15) \quad J_D \geq \log Z_* + 2(1 + \delta) \log(\beta - 1) - 2\delta \log \pi + 2\delta \log a - (1 + \delta) \log \mathcal{A}^*.$$

Interval evaluation encloses this lower-bound expression. Its upper endpoint need not bound J_D . For rational s and rational polynomial data all digamma shifts in κ_P reduce to rational corrections and the single constant

$$C_s = \frac{2}{2s-1} - 2(s-1)^2 \sum_{k \geq 1} \frac{1}{k(k+s-1)(k+2s-2)}.$$

The positive sum has tail at most $1/(2N^2)$ after N terms. Hence C_s can be enclosed using a finite sum and this explicit remainder. Together with the endpoint moments and (7.6), these bounds give the archimedean inputs to Proposition 6.1. Section 8 supplies the finite-place inputs and evaluates the resulting margin.

We also use the following digamma enclosure. For $x > 0$, integers $L \geq 0$ and $m \geq 1$, and $z = x + L$,

$$\left| \psi(x) - \left(\log z - \frac{1}{2z} - \sum_{k=1}^{m-1} \frac{B_{2k}}{2kz^{2k}} - \sum_{j=0}^{L-1} \frac{1}{x+j} \right) \right| \leq \frac{|B_{2m}|}{2mz^{2m}},$$

where B_{2k} are rational Bernoulli numbers. To verify the remainder, expand $(z^2 + t^2)^{-1}$ as a finite geometric sum in Binet's identity [27, Equation (5.9.15)],

$$\psi(z) = \log z - \frac{1}{2z} - 2 \int_0^\infty \frac{t}{(t^2 + z^2)(e^{2\pi t} - 1)} dt.$$

The residual has a fixed sign and is bounded by the next integrated term, $|B_{2m}|/(2mz^{2m})$. Apply the digamma recurrence L times. We evaluate the displayed expression with exact rational arguments and Bernoulli numbers, outward arithmetic, and the explicit remainder. The rational witness values are used without rounding the arguments.

8. THE RATIONAL WITNESS

We apply Proposition 6.1 with the following rational parameters and weights. Every parameter and weight written as a finite decimal is exact. The displayed inequalities are rounded outwards from more precise enclosures.

8.1. Parameters and Local Weights. Set

$$\delta = \frac{83647}{2000000}, \quad p = \frac{2}{1 + \delta}, \quad C = \frac{42161819}{10^9}, \quad \theta_* = \frac{4095}{8192}, \quad \varepsilon = 10^{-9}.$$

The real-place Gaussian is $f_C(z)^p = e^{-a_C|z|^2}$ with $a_C = 2p\delta$. At a complex pair, use the polynomial profile and coefficient matrix in (7.1), with $s = 1.14600585$ and $a = 0.0000128633241758$. For the Fourier estimate, take $\rho = 10^{-3}$. For the mass integral, use an 8×8 rational subdivision and binomial degree $N = 18$ (indices $0, \dots, 18$). For the overlap, retain orders zero and one with the signed remainder proved above.

At each selected rational prime r , let $Q = r^{f_r}$ and use identical six-shell profiles on the two local coordinates, with weights $w_{r,0}, \dots, w_{r,5}$. All $w_{r,0} = 1$. The absolute local indices and additive periods are

r	2	3	5	7	11	13	17	19	23	29	31
e_r	8	2	2	2	2	2	1	1	1	1	1
f_r	4	2	2	4	4	4	4	4	4	4	4
k_r	7	9	6	2	2	1	1	1	1	1	1

The weights in Tables 1 and 2 are strictly positive and nonincreasing. Rational comparisons verify these properties and positivity of the Bernstein coefficients. The finite local factor is

$$(8.1) \quad \log \mathcal{F}_{\delta,r} = -\delta k_r \log Q + \log((k_r + 1)B_r^2 + 2B_r R_r) - 2(1 + \delta) \log A_r,$$

where $A_r = \sum_i m_i w_{r,i}^p$, $B_r = \sum_i m_i w_{r,i}^2$, $R_r = \sum_{i,j} D_{ij} w_{r,i} w_{r,j}$, and the rational shell masses m_i and matrix D are those of Section 6. Expanding the shell weights into a nonnegative sum of ball indicators gives a second calculation of (8.1). The overlap numerators agree exactly as rational numbers.

TABLE 1. Exact weights for shells 1, 2 and 3. All $w_{r,0} = 1$.

r	$w_{r,1}$	$w_{r,2}$	$w_{r,3}$
2	$5.2903102784782 \cdot 10^{-2}$	$2.3157958402044 \cdot 10^{-3}$	$1.0125841840494 \cdot 10^{-4}$
3	$9.8866787345973 \cdot 10^{-2}$	$8.1985230138641 \cdot 10^{-3}$	$6.7798519183202 \cdot 10^{-4}$
5	$3.1389392203562 \cdot 10^{-2}$	$8.3331551426164 \cdot 10^{-4}$	$2.2242307460119 \cdot 10^{-5}$
7	$2.1902148915124 \cdot 10^{-4}$	$3.888831805136 \cdot 10^{-8}$	$8.3349312286089 \cdot 10^{-14}$
11	$2.2066794391725 \cdot 10^{-5}$	$5.2893244151477 \cdot 10^{-10}$	$3.2351541879579 \cdot 10^{-16}$
13	$1.9753217475271 \cdot 10^{-5}$	$4.7825791524842 \cdot 10^{-11}$	$1.154930945714 \cdot 10^{-16}$
17	$5.1855578363751 \cdot 10^{-6}$	$4.661704085405 \cdot 10^{-12}$	$4.1907708339914 \cdot 10^{-18}$
19	$3.0150737786821 \cdot 10^{-6}$	$1.7755448176562 \cdot 10^{-12}$	$1.0455994096796 \cdot 10^{-18}$
23	$1.2033271761207 \cdot 10^{-6}$	$3.17236260693 \cdot 10^{-13}$	$8.3633817213758 \cdot 10^{-20}$
29	$4.0183950841977 \cdot 10^{-7}$	$4.2255710985501 \cdot 10^{-14}$	$4.4434284649605 \cdot 10^{-21}$
31	$2.9331492202408 \cdot 10^{-7}$	$2.5364272697667 \cdot 10^{-14}$	$4.0395513941639 \cdot 10^{-21}$

8.2. **Certified Enclosures.** Let $q = sp - 1$, and let T, U be independent with density $qt^{q-1} dt$ on $(0, 1)$. The mass and overlap formulas of Section 7 give

$$(8.2) \quad 38.9687816563247549 < \mathbb{E}[P(T, U)^p] < 38.9687816563247561,$$

$$(8.3) \quad 348.4186894704059951 < Z_* < 348.4186894704059952.$$

TABLE 2. Exact weights for shells 4 and 5. All $w_{r,0} = 1$.

r	$w_{r,4}$	$w_{r,5}$
2	$4.4907846711318 \cdot 10^{-6}$	$2.0159593157185 \cdot 10^{-7}$
3	$5.6589534354296 \cdot 10^{-5}$	$4.7456521092252 \cdot 10^{-6}$
5	$6.0504358878881 \cdot 10^{-7}$	$1.6737711273851 \cdot 10^{-8}$
7	$1.6239748423571 \cdot 10^{-18}$	$3.1643053340993 \cdot 10^{-23}$
11	$1.2374208229268 \cdot 10^{-21}$	$4.7330769061366 \cdot 10^{-27}$
13	$2.7890532331439 \cdot 10^{-22}$	$6.7353099910173 \cdot 10^{-28}$
17	$3.7674120582188 \cdot 10^{-24}$	$3.3868217038473 \cdot 10^{-30}$
19	$6.1574234266266 \cdot 10^{-25}$	$3.6260409965598 \cdot 10^{-31}$
23	$2.2048599887241 \cdot 10^{-26}$	$5.812729505639 \cdot 10^{-33}$
29	$4.6725178828482 \cdot 10^{-28}$	$4.9134184420207 \cdot 10^{-35}$
31	$6.4334537606741 \cdot 10^{-28}$	$1.0246020721641 \cdot 10^{-34}$

Here Z_* is the lower expression for the overlap used in (7.15). The upper endpoint in (8.3) bounds this expression and gives no upper bound for the overlap integral. Write J_D^* for the right side of (7.15) with the certified mass allowance. Thus $J_D \geq J_D^*$. The finite terms satisfy

$$1.0335669225039939710 < \sum_r \frac{\log \mathcal{F}_{\delta,r}}{e_r f_r} < 1.0335669225039939712.$$

The rational polynomial tube bound satisfies $q_0 < 0.026 < 1$. The Fourier and period calculation yields

$$\log K_D < 0.04916819, \quad \sigma_D > 1.7518755, \quad \mu > 26497.47.$$

Here $\sigma_D = 2\pi\rho/\sqrt{a}$, and K_D is the Fourier prefactor in (7.6). Together with the Gaussian bound, these inequalities imply (6.7). The endpoint moments are finite and the common energy window is compact by Sections 5, 6 and 7.

Define the conservative margin

$$\begin{aligned} M_*(\theta) = & \sum_r \frac{\log \mathcal{F}_{\delta,r}}{e_r f_r} - (1/2 - \delta)\ell - C + (1 - \delta) \log 2 + (1 - \theta) \log \pi \\ & + (1 - 2\theta)J_C + \theta J_D^*. \end{aligned}$$

Since $\theta \geq 0$, this is a lower bound on (6.10). Its signature slope is $-\log \pi - 2J_C + J_D^* > 0.6492390240$. It suffices to evaluate the margin at the smallest permitted signature. Directed interval arithmetic gives

$$(8.4) \quad M_*(\theta_*) > 0.00000533604359816794,$$

$$(8.5) \quad M_*(\theta_*) - 4\varepsilon > 0.00000533204359816794 > 0.$$

These evaluations use the exact inputs and full interval expressions. The abbreviated decimal bounds displayed here are not substituted into later calculations.

Proof of Theorem 1.1. Theorem 2.6 gives finite Galois fields K_j of unbounded degree with the prescribed root discriminant and local indices. Their actual complex conjugations define quadratic extensions K_j/F_j unramified at every finite place, with all selected primes split and $\theta_j \geq \theta_*$. Theorem 4.1 gives $d_j^{-1} \log L_{F_j}(1) < C$ for all sufficiently large j .

The profiles above satisfy the positivity, period, moment and Fourier conditions of Proposition 6.1. Their margin is at least $M_*(\theta_*)$, and (8.5) is positive. The proposition gives finite planar sets whose unordered edge ratio and cardinalities tend to infinity, proving Theorem 1.1. $\square$

8.3. The Supplementary Certificate. The rational witness is `certificates/witness.json`. The source and arithmetic data are distributed as the three parts `certificates/data.tar.gz`, `part00` through `part02`. The replay script verifies and joins the parts. The part and member SHA-256 index is `certificates/data-manifest.json`. The archive includes the coefficient generators and complete retained moment tables, actual-field arithmetic receipts, and the prime census through 10^{12} . Its internal directory names record computational provenance. The proofs are given in this manuscript.

Finite-field linear algebra verifies the quadratic forms, ranks and representation counts. Exact Euler recurrences enumerate signed and absolute coefficients. Integer bin moments, the explicit kernel remainders, and outward arithmetic enclose the Hecke values. A prime sieve supplies the finite Frobenius correction. Rational polynomial operations and the mass and overlap remainders of Section 7 enclose the geometric score. The preceding sections justify these calculations. The verifier evaluates the fixed witness and contains no optimizer.

The replay works on a private copy of the supplied inputs, which it leaves unchanged, and has three levels. The basic level checks the retained input identities and recomputes the final geometric witness. The analytic level also evaluates every Hecke factor at 12001/12000, checks the independent prime-census prefix and the full finite correction, and reassembles the Euler ceiling. The moment level also regenerates the 64 complete sector files of the five newest sparse producers and compares every integer bin; these contain 103012 bins and 5515960 integer moments. Earlier moment tables and actual-field arithmetic receipts remain supplied finite data at every level, and no level repeats the full 10^{12} prime sieve, whose source is included.

The arithmetic software uses PARI/GP for exact field and Euler computations, Arb for ball arithmetic, and MPFR and mpmath intervals for directed enclosures [40, 16, 38, 39]. The package README gives the replay commands, dependency versions and build instructions. A manifest binds the bytes of every source file and the bibliography of this manuscript together with the certificate archive, and its checker rejects any change to that set. These checks establish source identity. The mathematical proofs and the finite computations remain separate obligations.

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